simon-5502-06-slides

Topics to be covered

- What you will learn
  - A humorous look at nonlinear fits
  - Constructing a spline from small pieces
  - B-splines
  - Placement and number of knots
  - Splines for linear and logistic regression
  - Your homework

Web page on overfitting

- Simon, Overfitting example using hurricane frequencies, PMean blog 2021-12-11, available in html format

06-01 xkcd comic

- Title “CURVE-FITTING METHODS AND THE MESSAGE THEY SEND”
- Drawn by Scott Munro
- Open-source license
- Link to comic at xkcd.com
- More details at explain-xkcd.com

Speaker notes

Let me start with a cartoon from the xkcd site of Scott Munro. Scott Munro produces comics that poke fun at various scientific and mathematical concepts. There are a handful that directly address statistics, including this one. The actual panels in the comic are small, so I split them up onto separate slides.

06-01 xkcd comic, panel 01

Speaker notes

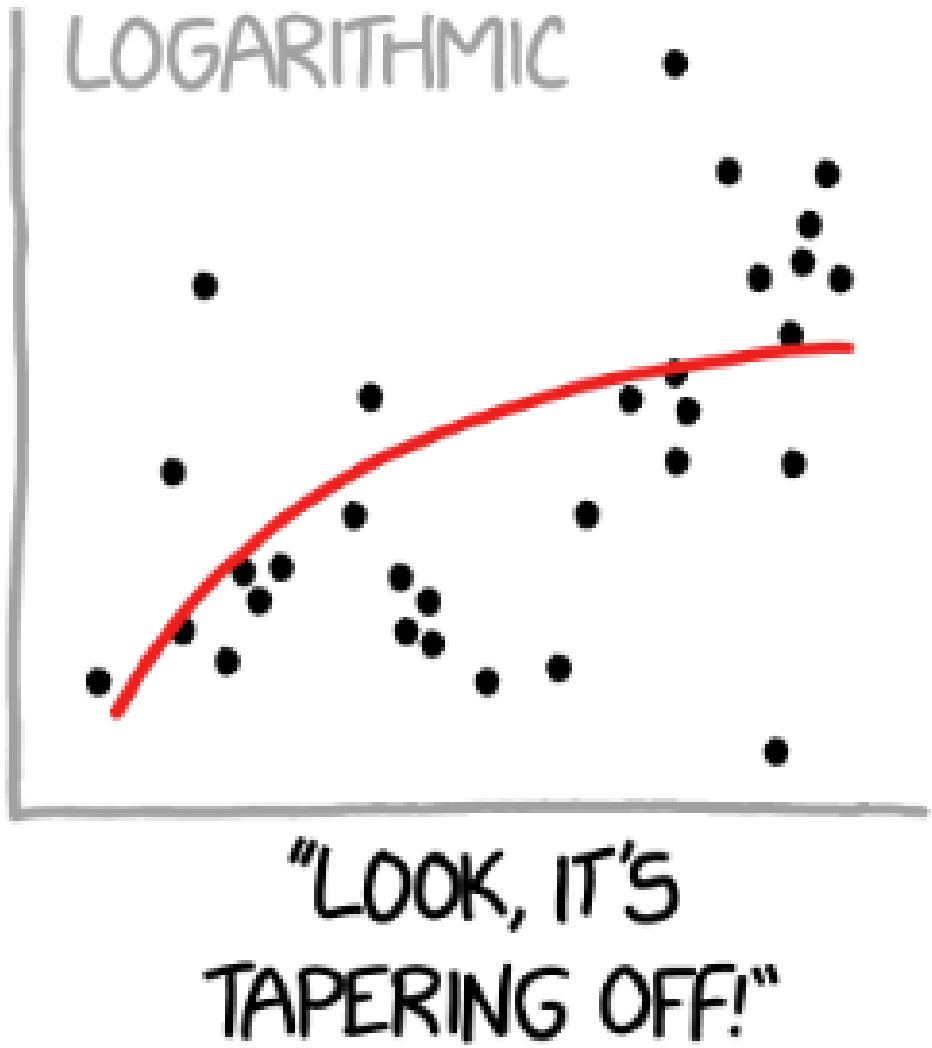
This shows a scatter of data points with a linear regression fit. The caption reads, “Hey, I did a regression.”

06-01 xkcd comic, panel 02

Speaker notes

This shows a scatter of data points with a quadratic regression fit. The caption reads, “I wanted a curved line, so I made one with Math.”

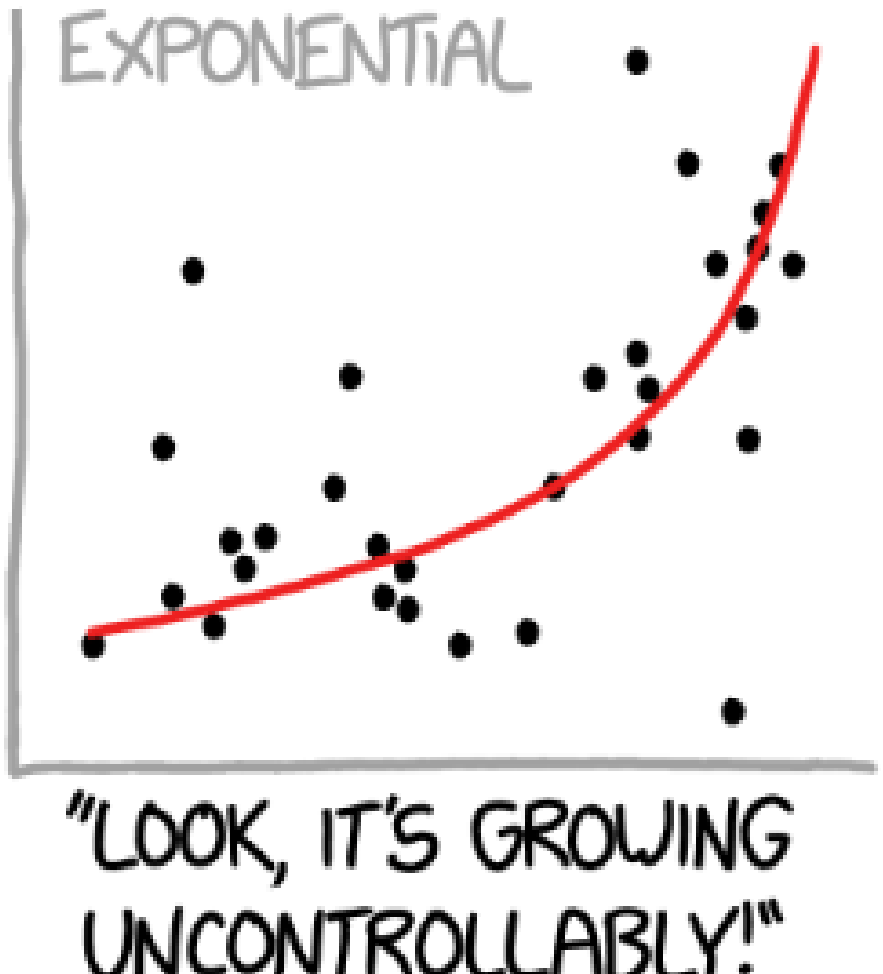
06-01 xkcd comic, panel 03



Speaker notes

This shows a scatter of data points with a logarithmic regression fit. The caption reads, “Look, it’s tapering off!”

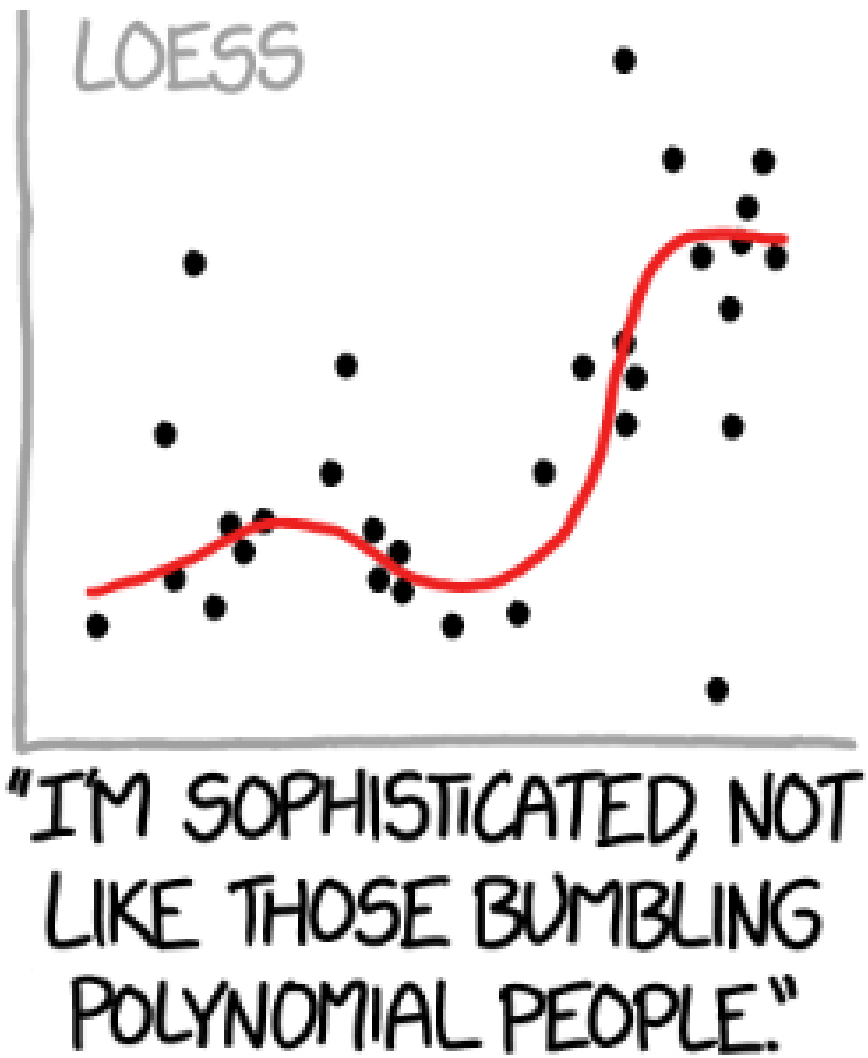
06-01 xkcd comic, panel 04



Speaker notes

This shows a scatter of data points with an exponential regression fit. The caption reads, “Look, it’s growing uncontrollably.”

06-01 xkcd comic, panel 05



Speaker notes

This shows a scatter of data points with a loess smoothing curve. The caption reads, “I’m sophisticated, not like those bumbling polynomial people.”

06-01 xkcd comic, panel 06

Speaker notes

This shows a scatter of data points with a flat linear regression fit (slope=0). The caption reads, “I’m making a scatter plot but I don’t want to.”

06-01 xkcd comic, panel 07

Speaker notes

This shows a scatter of data points with a logistic curve regression fit. The caption reads, “I need to connect these two lines, but my first idea didn’t have enough Math.”

06-01 xkcd comic, panel 08

Speaker notes

This shows a scatter of data points with no particular regression line, but a very wide (nd probably appropriate) confidence band. This captures the idea that there is uncertainty not only in the deviation of the points from the regression curve, but true uncertainty about the shape of that regression curve. The caption reads, “Listen, science is hard. But I’m a serious person doing my best.”

There is an active field of research under the topic uncertainty quantification that tries to take into account all the sources of uncertainty including uncertainty about which model is the correct model.

06-01 xkcd comic, panel 09

Speaker notes

This shows a scatter of data points with a piecewise linear regression fit. The caption reads, “I have a theory and this is the only data I could find.”

06-01 xkcd comic, panel 10

Speaker notes

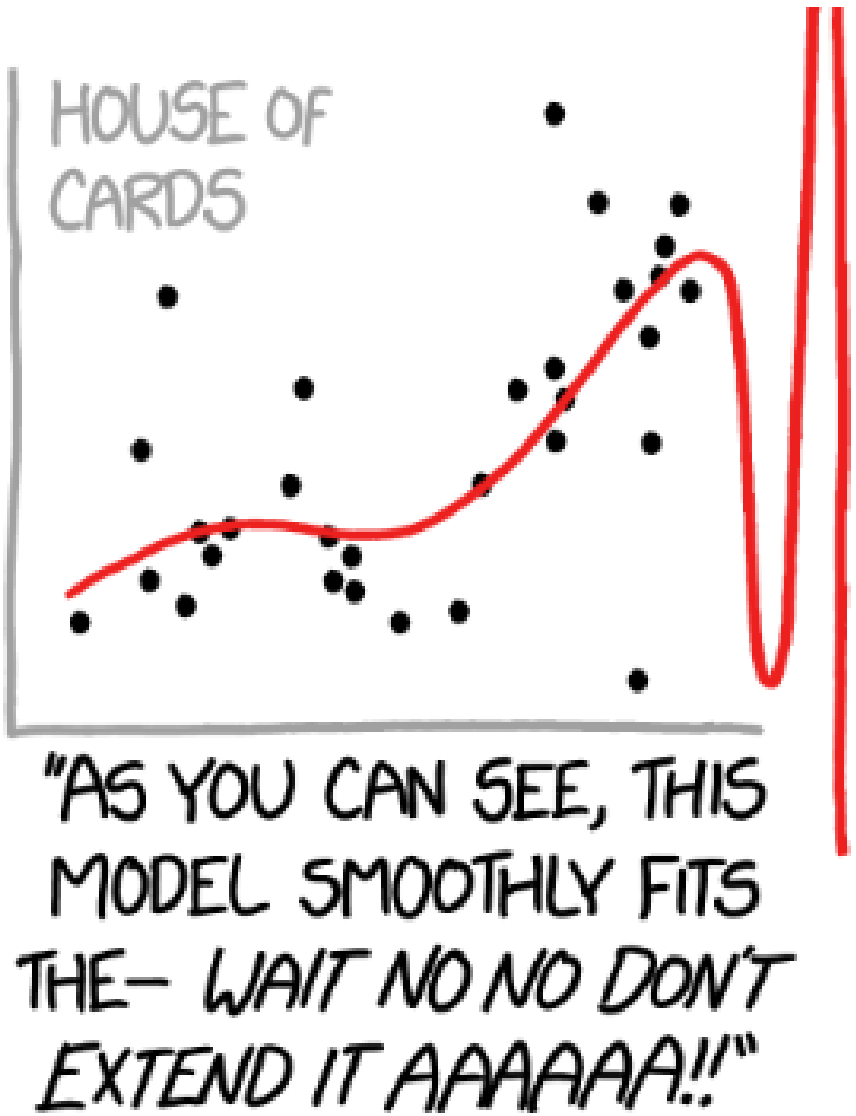
Here the proposed regression model gets a bit silly. This shows a scatter of data points with a smooth curve connecting every data point. The caption reads, “I clikced ‘smooth lines’ in Excel.”

06-01 xkcd comic, panel 11

Speaker notes

This shows a scatter of data points with a smoother that looks like it uses medians somehow. The caption reads, “I had an idea for how to clean up the data. What do you think?”

06-01 xkcd comic, panel 12



Speaker notes

This shows a scatter of data points with what looks like a B-spline. The caption reads, “As you can see, this model smoothly fits the - Wait, No, No, Don’t extend it. AAAAAA!!”

06-01 Examples of splines, 1

Current Research in Microbial Sciences 9 (2025) 100412

Contents lists available at ScienceDirect

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journal homepage: www.sciencedirect.com/journal/current-research-in-microbial-sciences

Association between dietary index for gut microbiota and sleep duration in US adults: a cross-sectional study

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ARTICLE INFO

Keywords:
Dietary index for gut microbiota
DI-GM
NHANES
Sleep duration

ABSTRACT

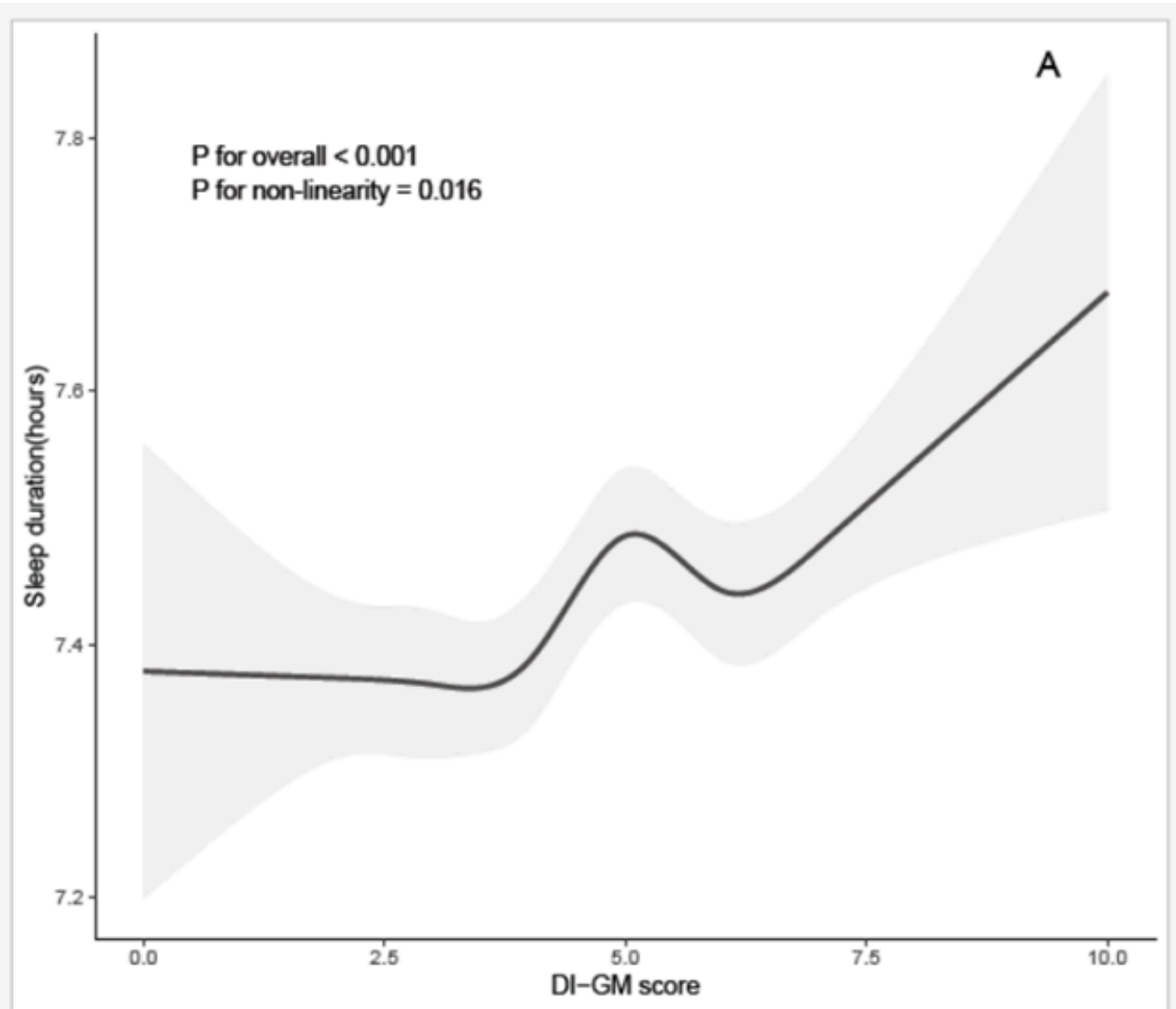
Background: The microbiota-gut-brain axis underlies the pathogenesis of sleep disorders. The dietary index for gut microbiota (DI-GM) is an innovative diet quality metric related to gut microbiota diversity, with higher scores indicating a healthier gut microbiome. This study aimed to investigate its association with sleep duration. *Methods:* This cross-sectional study analyzed the National Health and Nutrition Examination Survey (NHANES) data from 2005 to 2018, involving 25,439 participants aged ≥20 years. The DI-GM score was calculated utilizing dietary recall data, encompassing both beneficial (BGMS) and unfavorable (UGMS) to gut microbiota score. Sleep

Speaker notes

Here’s a peer-reviewed study of sleep duration and how it relates to your gut microbiota–all those tiny bugs living inside your intestines. The authors used a composite score called the dietary index for gut micrtobiota with the acronym DI-GM. This article is published under an open source license, like all the other papers you will see today.

Liu J, Huang S. Association between dietary index for gut microbiota and sleep duration in US adults: a cross-sectional study. Curr Res Microb Sci. 2025 May 27;9:100412. doi: 10.1016/j.crmicr.2025.100412. PMID: 40528895; PMCID: PMC12171763.

06-01 Examples of splines, 2



Speaker notes

The cubic spline starts out flat for low values of DI-GM, then rises, dips, and rises again.

06-01 Examples of splines, 3

Wang et al. Lipids in Health and Disease (2025) 24:209
https://doi.org/10.1186/s12944-025-02628-0

Lipids in Health and Disease

RESEARCH Open Access

Association between triglycerides and remnant cholesterol levels and spine bone mineral density in Duchenne muscular dystrophy

Yayu Wang[1,2], Yue Chang[1,2], Pei Zhang[1], Zhuoyin Zheng[1,3], Xiuyi Ai[1,2], Shu Zhang[1*] and Shiwen Wu[1,2*]

Abstract

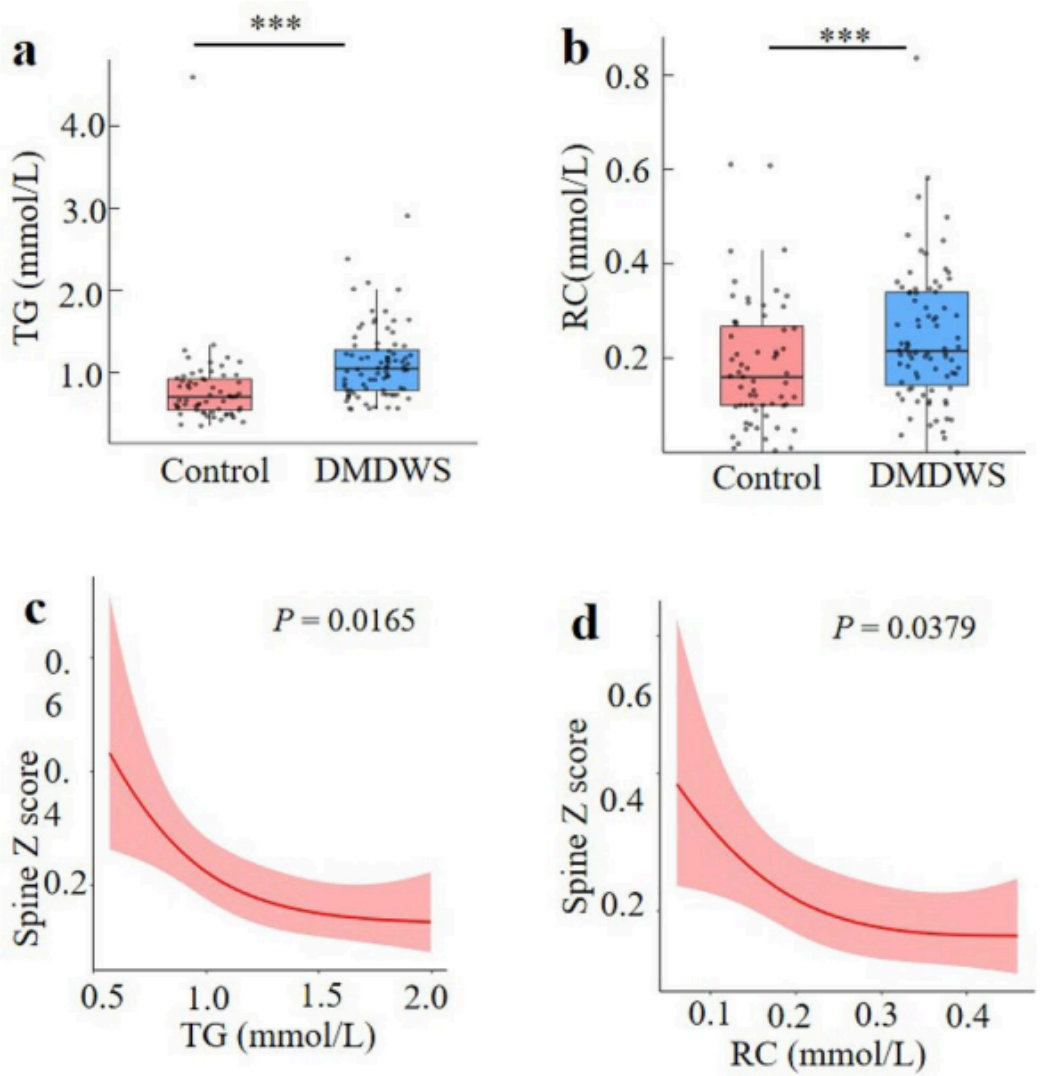
Background Duchenne muscular dystrophy (DMD) is a serious, progressive neuromuscular condition that predominantly impacts male individuals, marked by progressive muscle weakness resulting from mutations in the dystrophin gene *(DMD)* encoding dystrophin. DMD is a primary muscle disorder that often presents with secondary abnormalities in lipid metabolism and decreased bone mineral density. Although disturbances in circulating lipid profiles and skeletal health have been observed in individuals with DMD, their relationship remains underexplored.

Speaker notes

Here’s another open source article, a peer-reviewed study of patients with Duchene muscular dystrophy (DMD). The authors measured spine bone mineral density and sought to see how that related to various cholesterol measures, including triglycerides and remnant cholesterol both in patients with DMD and age-matched healthy controls. The authors controlled for a variety of factors including steroid use.

Wang Y, Chang Y, Zhang P, Zheng Z, Ai X, Zhang S, Wu S. Association between triglycerides and remnant cholesterol levels and spine bone mineral density in Duchenne muscular dystrophy. Lipids Health Dis. 2025 Jun 9;24(1):209. doi: 10.1186/s12944-025-02628-0. Erratum in: Lipids Health Dis. 2025 Jun 27;24(1):222. doi: 10.1186/s12944-025-02646-y. PMID: 40490739; PMCID: PMC12147359.

06-01 Examples of splines, 4



Speaker notes

The box plot examines DMD patients without corticosteroid treatment (DMDWS) to the healthy controls. Triglycerides and remnant cholesterol was elevated in the DMDWS group. The cubic spline shows a deline in bone health for larger values of triglycerides and remnant cholesterol, but the effect is not quite as strong for larger values of these cholesterol measurements.

06-01 Examples of splines, 5

frontiers | Frontiers in Nutrition

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Hu C and Tang T (2025) Association between

Association between niacin intake and chronic kidney disease in male participants—a cross-sectional study from the NHANES (2005–2018)

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Department of Traditional Chinese Medicine, The Central Hospital of Wuhan, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei, China

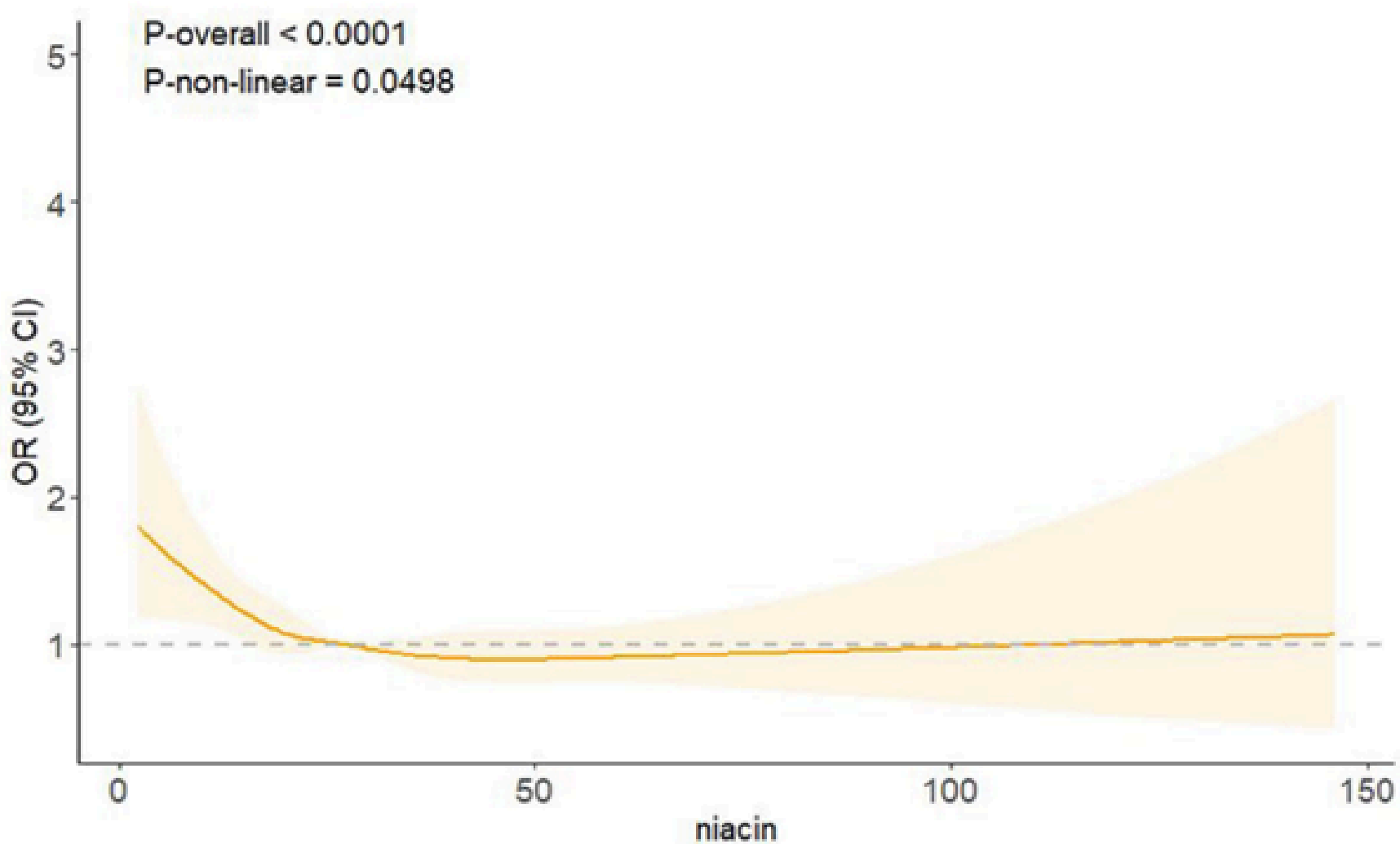
Background: Chronic kidney disease (CKD), a significant health challenge in the United States, often progresses from asymptomatic conditions to advanced stages and exhibits a higher prevalence among male individuals. Niacin is known for its metabolic and antioxidant roles, potentially influencing CKD progression. The association between niacin intake and CKD has been rarely investigated in

Speaker notes

Here’s a peer-reviewed study of how nutrtion is associated with the risk of chronic kidney disease (CKD). The authors got their data from the National Health and Nutrition Examination Survey (NHANES) data, a large scale CDC survey that includes a medical exam and dietary history. They estimated niacin intake through a dietary recall and classified patients as having CKD based on their Albumin to creatinine ratio and the estimated glomerular filtration rate.

Hu C, Tang T. Association between niacin intake and chronic kidney disease in male participants-a cross-sectional study from the NHANES (2005-2018). Front Nutr. 2025 Jun 13;12:1578118. doi: 10.3389/fnut.2025.1578118. PMID: 40584108; PMCID: PMC12202428.

06-01 Examples of splines, 6



Speaker notes

The cubic spline shows that a deficiency of niacin increases the odds of CKD, but that larger levels of NIACIN above 20 to 30 do not have an impact.

06-01 A real-world problem, without the data

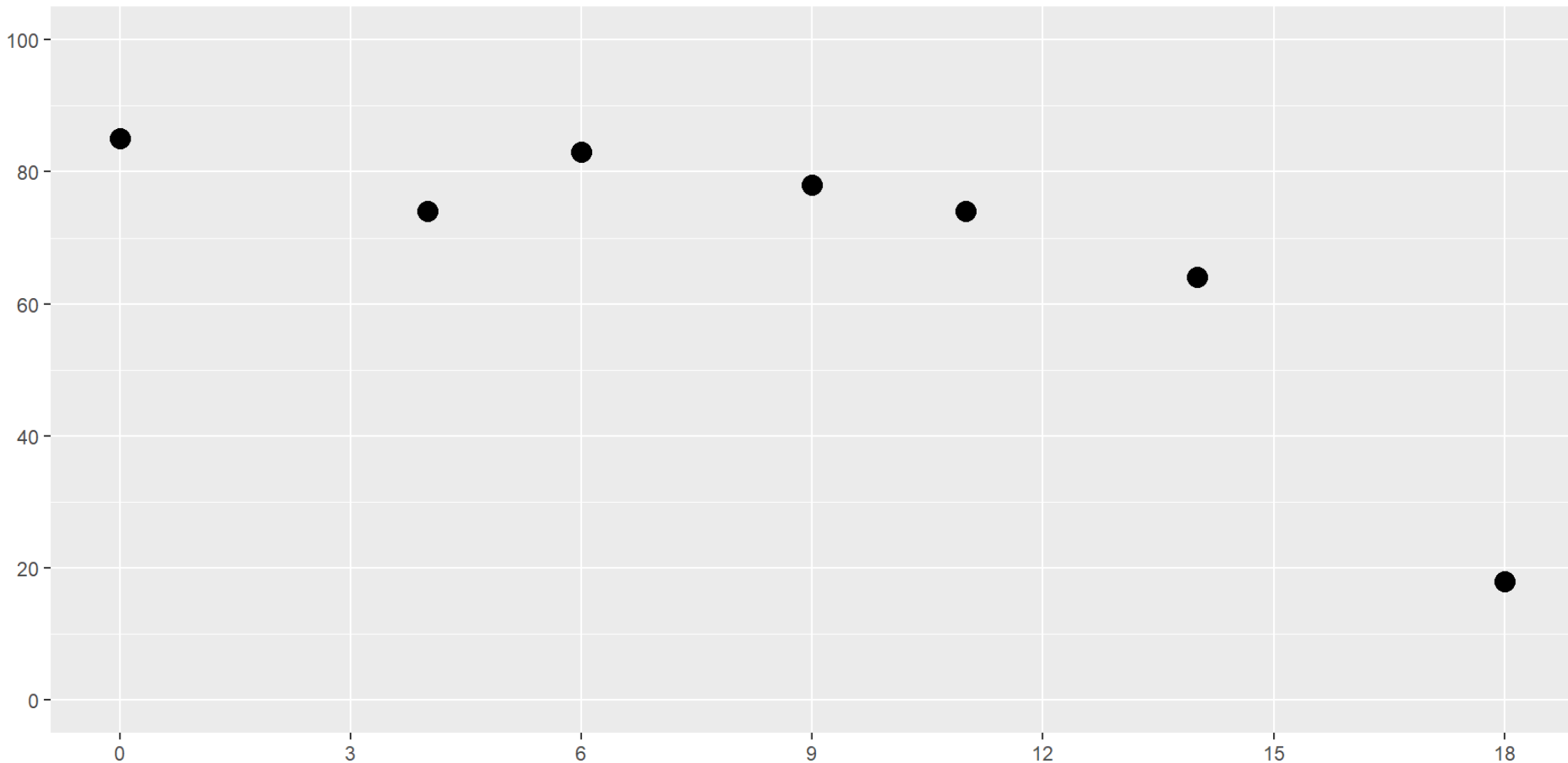
- Threshold model
  - Nothing happens until you meet a threshold
  - Then things get worse

Speaker notes

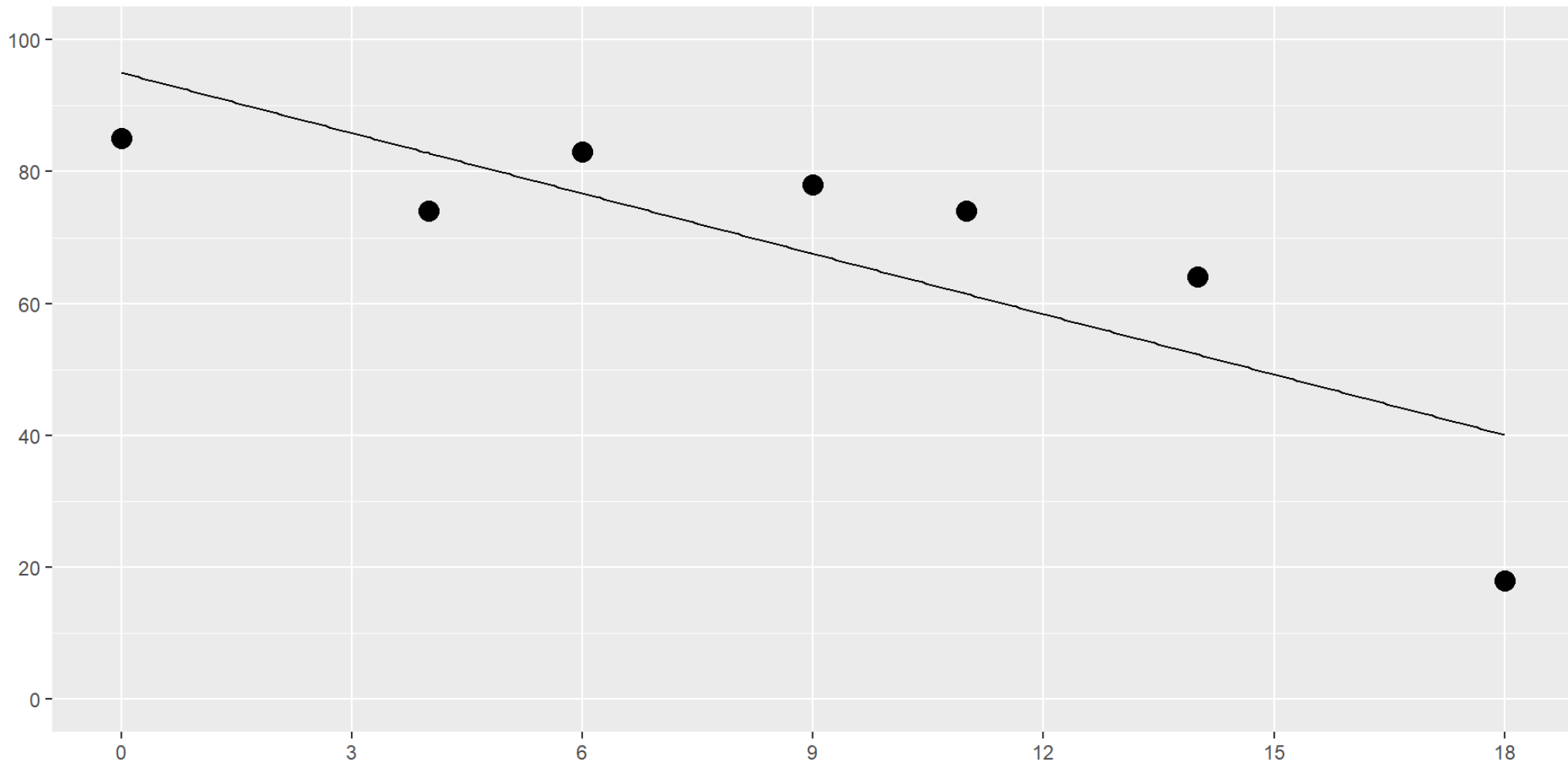
Several decades ago, I was faced with a data analysis problem. I wanted to fit a threshold model where everything is fine and normal until the exposure level meets a certain threshold. Then things get worse.

I don’t have the data for this problem but let me illustrate conceptually how a threshold model might work. It provides a simple but useful analog to regression splines.

06-01 Fake data



06-01 Linear function



Speaker notes

Here is what a linear function might look like. It is not what we want, but always start with the easiest model.

06-01 Computational formula for linear regression

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 4 \\ 1 & 6 \\ 1 & 9 \\ 1 & 11 \\ 1 & 14 \\ 1 & 18 \end{bmatrix}$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$

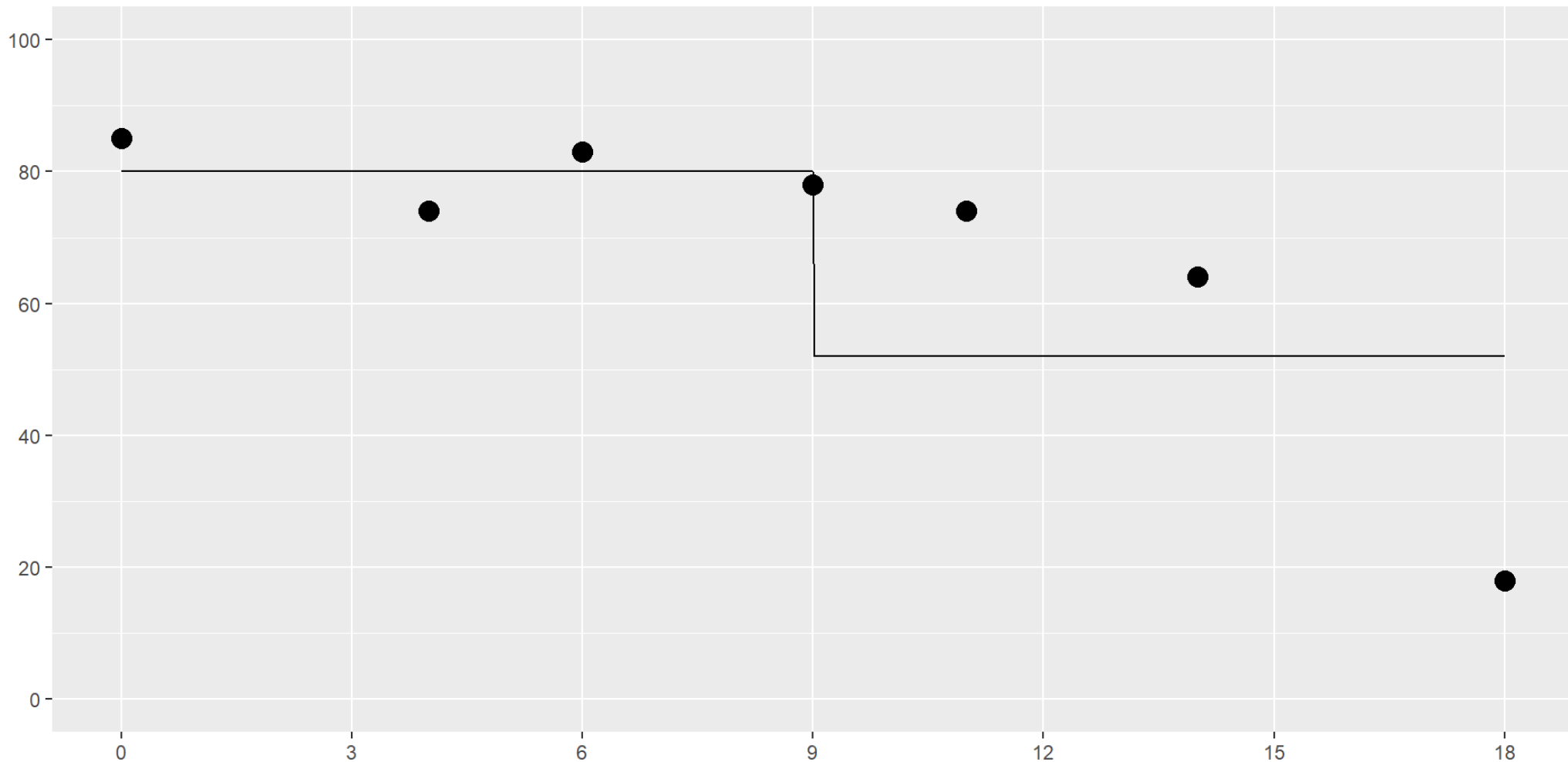
Speaker notes

This is the matrix formula for linear regression. Don’t worry if you haven’t used matrices before. Just focus on how the X matrix is laid out for linear regression. The first column is all 1’s and the second column contains the X values.

In multiple linear regression, you would have additional columns, but let’s keep it simple for now.

I do want to note here that you can get a different regression model by changing the second column.

06-01 Step function



Speaker notes

Here is what a step function might look like. It is better than a linear fit, but the sudden jump at x=9 is not quite what you want.

This is a discontinuous function. It might make sense in some settings. In other settings, you might expect the decline to be not so sudden and abrupt.

06-01 Computational formula for step regression

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$

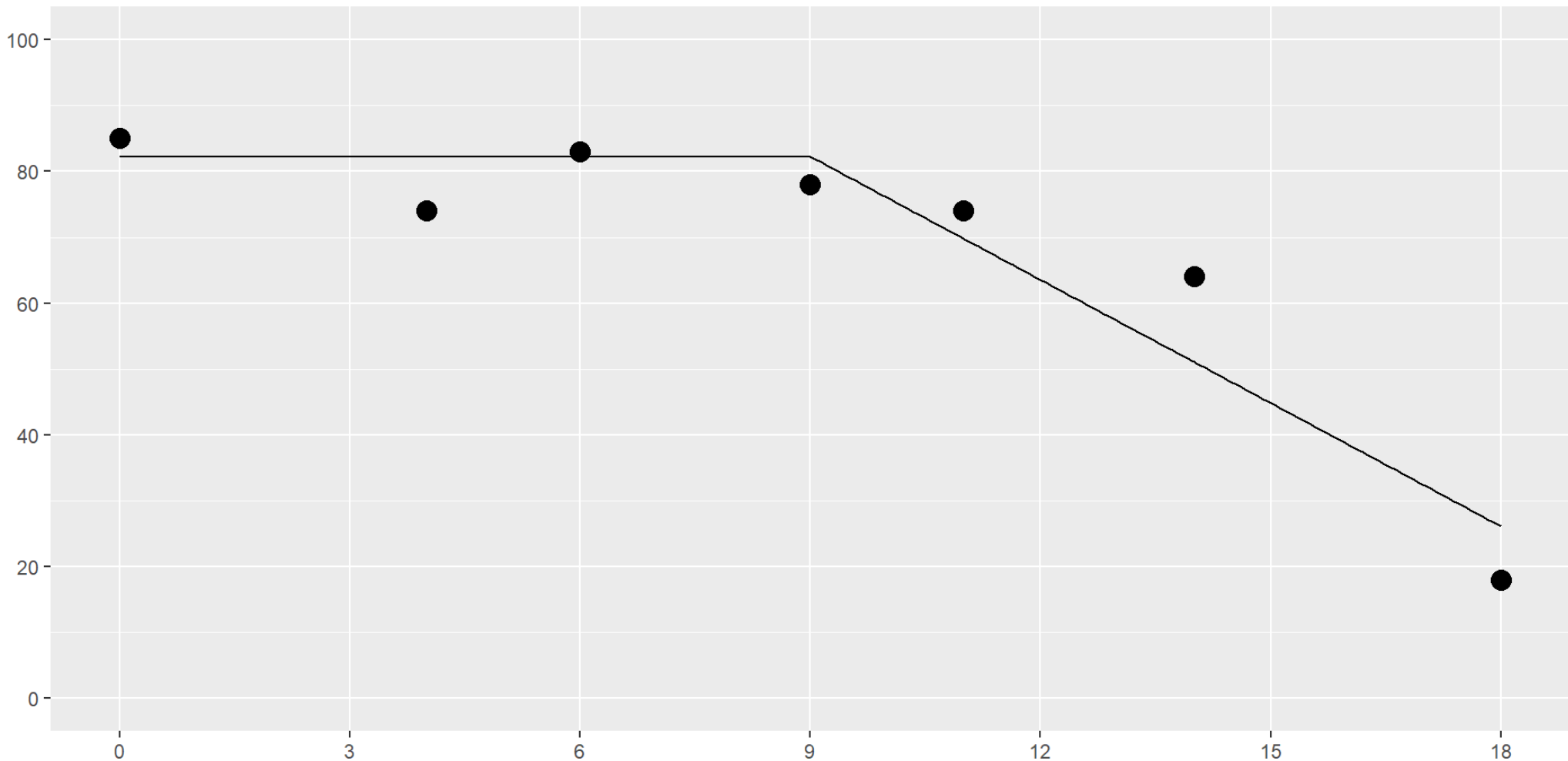
Speaker notes

This is the matrix you would use if you wanted a step regression model. The first column is all 1's and the second column is zero for X values of 9 or less and 1 for X values larger than 9.

Notice that the matrix formula has not changed. You are familiar with all the mathematical properties for linear regression. You know how to get confidence intervals and hypothesis tests for linear regression. The step regression model will use pretty much the same formulas, just with a slight tweak to the X matrix.

More importantly, any statistical software that can compute a linear regression can also compute a step regression.

06-01 Elbow function



Speaker notes

A threshold model could look like this. It is a step function that is at a high level prior to the threshold at X=9 and then drops after.

Here is a better model. It is flat and at a high level for values less than 9 and declines linearly for values greater than 9.

Note the “elbow” at the threshold. This might be okay, but it might be better for a transition that does not change slope so suddenly.

06-01 Computational formula for elbow regression

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 1 & 5 \\ 1 & 9 \end{bmatrix}$$

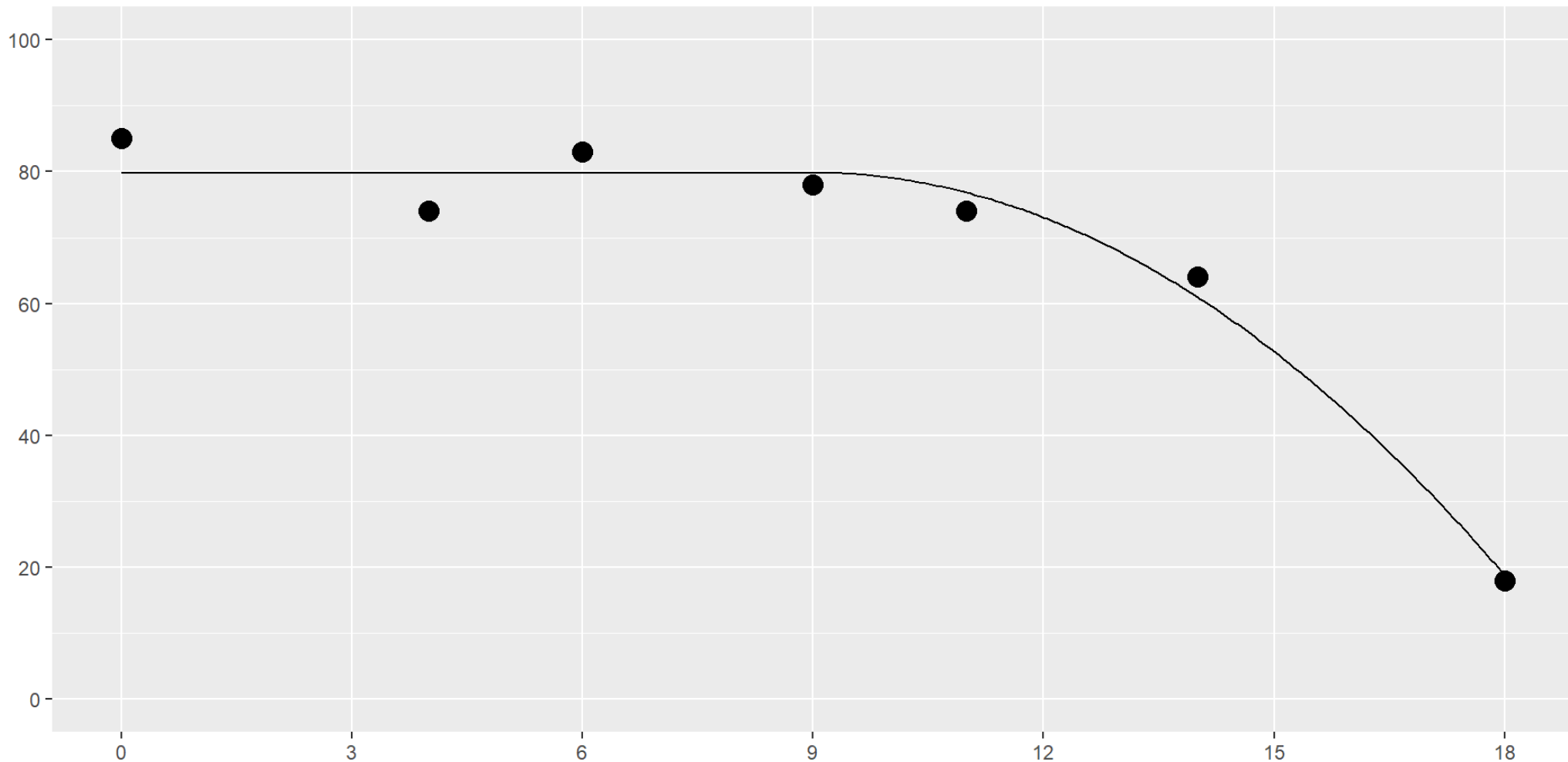
$$\hat{\beta} = (X'X)^{-1}X'Y$$

Speaker notes

This is the matrix you would use if you wanted the elbow regression. The first column is all 1's and the second column is zero for X values of 9 or less and (x-9) for X values larger than 9. Again, it is important to subtract the 9 because you want the linear section to just start descending at when X reaches 9.

Notice that the matrix formula still has not changed. Fitting an elbow regression is just as easy as a linear regression. You can do all the fun stuff and you can use the same software.

06-01 Quadratic analog



Speaker notes

Here is a functional form that avoid the elbow at the threshold value. It fits a quadratic decline, or a parabola, but the parabola starts at the point where the derivative is equal to zero.

06-01 Computational formula for quadratic analog

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 4 \\ 1 & 25 \\ 1 & 81 \end{bmatrix}$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$

Speaker notes

This is the matrix you would use if you wanted the quadratic analog to the elbow regression. The first column is all 1's and the second column is zero for X values of 9 or less and (x-9) squared for X values larger than 9. Again, it is important to subtract the 9 before squaring, because you want the quadratic curve to just start descending at when X reaches 9.

Notice that the matrix formula still has not changed. Fitting an elbow regression is just as easy as a linear regression. You can do all the fun stuff and you can use the same software.

Break #1

- What you have learned
  - A humorous look at nonlinear fits
- What’s coming next
  - Constructing a spline from small pieces

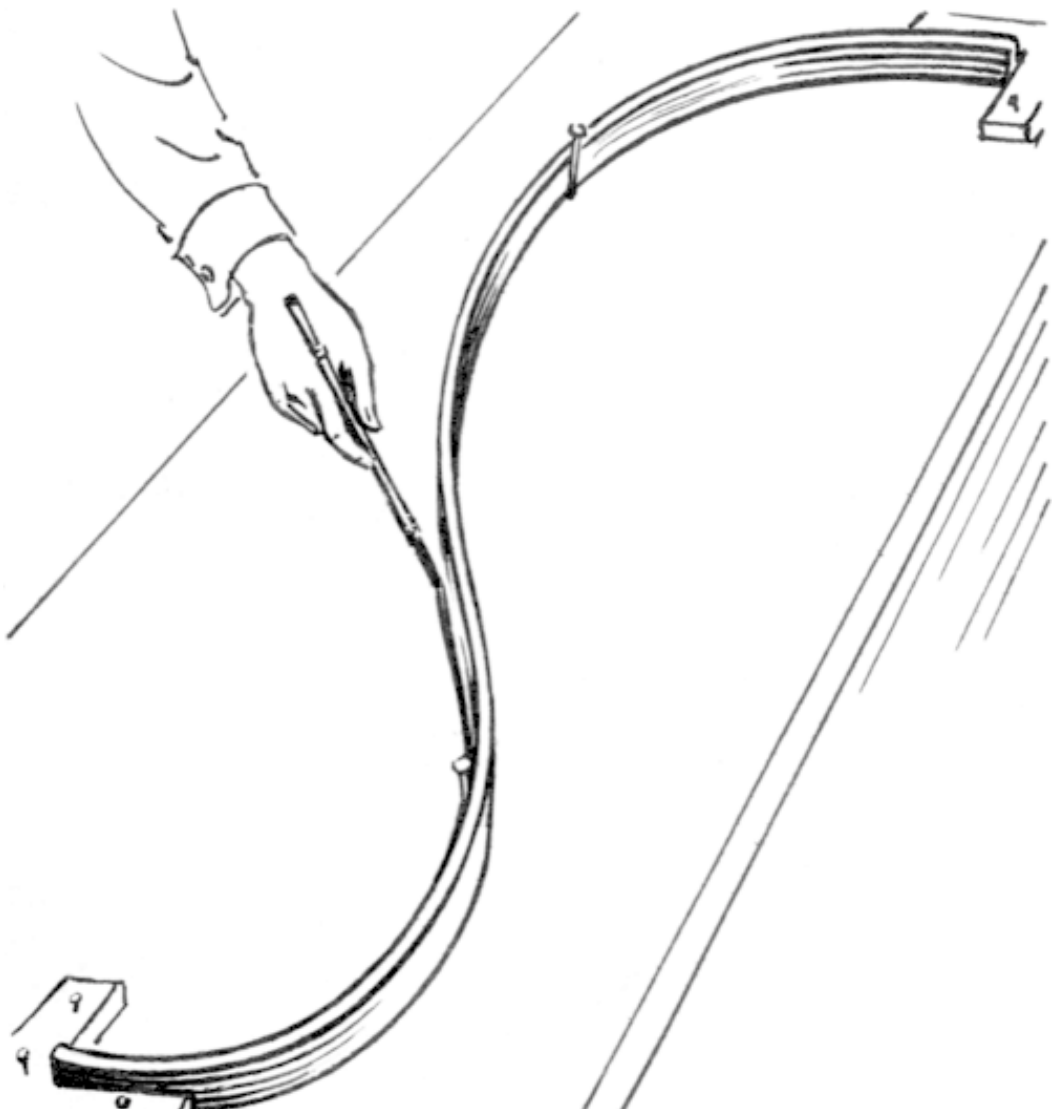
06-02 Splines

- Piecewise cubic polynomial
  - Continuous (no jumps)
  - Smooth (no sharp turns)
- Transition points = knots

Speaker notes

A spline is a piecewise cubic polynomial. It converts from one cubic polynomial to a different cubic polynomial at pre-defined transition points known as knots. The individual cubic polynomials are constrained to be continuous and smooth at the knots. Continuous means no sudden jumps. Smooth means no sudden turns. You can also think of it as no elbows. Mathematically, smooth means a continuous first and second derivative.

06-02 A physical spline



Speaker notes

Before computers became powerful enough to compute splines, we had several alternatives.

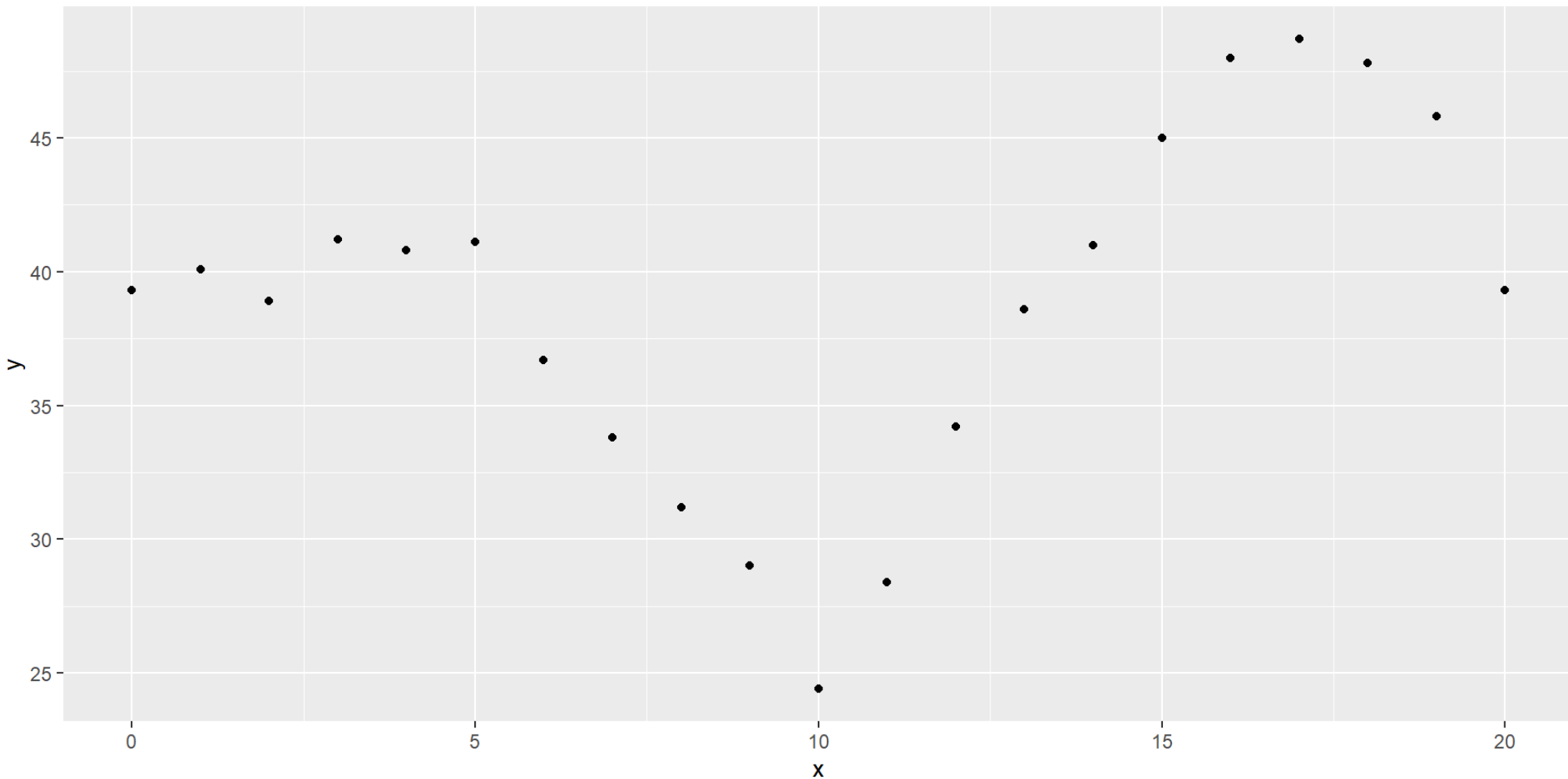
Here is an image from Wikipedia of physical spline. It is a thin strip of wood–thin enough to allow it to bend. It is constrained so that the bend follows a path that covers a few points.

06-02 A french curve

Speaker notes

The french curve was a plastic template that showed a range of curvatures. You would line up the curve to four data points, draw along the french curve between the two interior points and then shift to the right and line up the french curve again. The resulting path looked seamless.

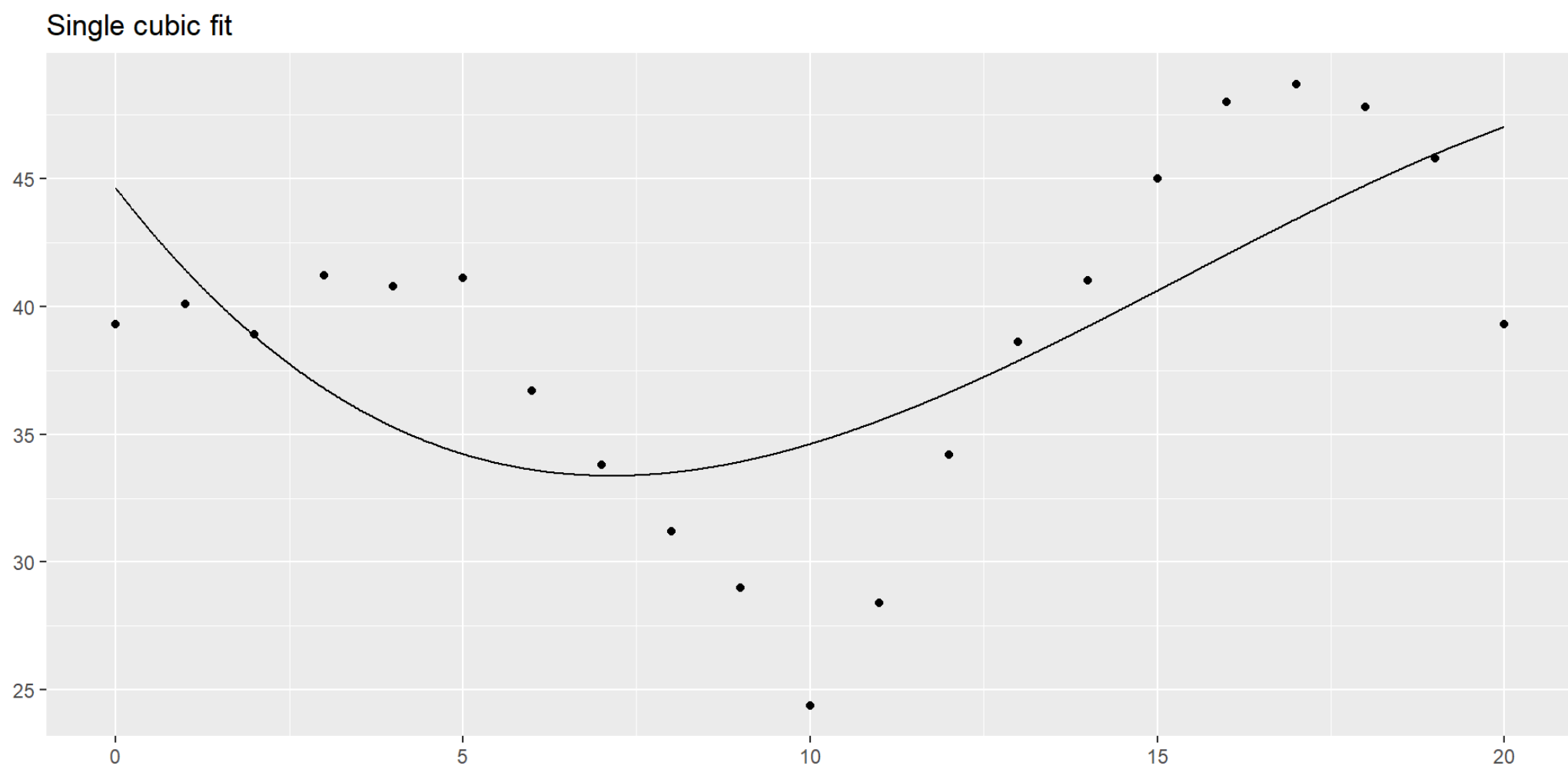
06-02 Graph of artificial data



Speaker notes

Here is some artificial data. I manipulated the data so that the behavior of the underlying trendfor x=0 to 5, x=6 to 10, x=11 to 15, and x=16 to 20.

06-02 Single cubic polynomial

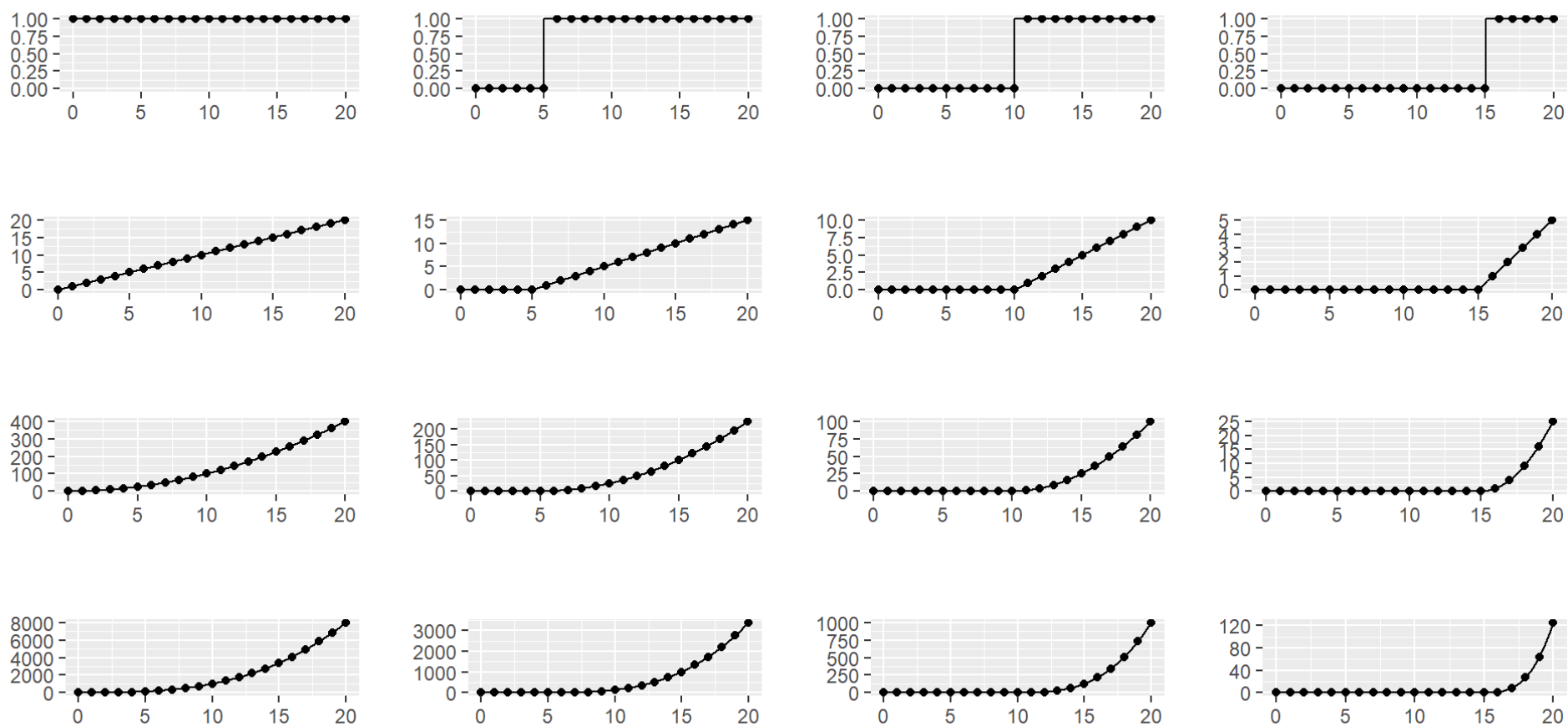


Speaker notes

The graph shown here represents the best fitting single cubic polynomial. It doesn’t fit the data very well. The problem is that the data is largely flat from 0 to 5 and then starts wiggling a lot from 5 onward. A cubic polynomial (or any polynomial for that matter) can be mostly flat or very wiggly but it generally can’t be both.

You need a way to transition between a wiggly cubic and a mostly flat cubic.

06-02 Cubic polynomials with knots at x=5, 10, 15



06-02 What the numbers look like

Overall Intercept	Overall Slope	Overall Quadratic	Overall Cubic	Knot 1 Intercept	Knot 1 Slope	Knot 1 Quadratic	Knot 1 Cubic	Knot 2 Intercept	Knot 2 Slope	Knot 2 Quadratic	Knot 2 Cubic	Knot 3 Intercept	Knot 3 Slope	Knot 3 Quadratic	Knot 3 Cubic
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	2	4	8	0	0	0	0	0	0	0	0	0	0	0	0
1	3	9	27	0	0	0	0	0	0	0	0	0	0	0	0
1	4	16	64	0	0	0	0	0	0	0	0	0	0	0	0
1	5	25	125	1	0	0	0	0	0	0	0	0	0	0	0
1	6	36	216	1	1	1	1	0	0	0	0	0	0	0	0
1	7	49	343	1	2	4	8	0	0	0	0	0	0	0	0
1	8	64	512	1	3	9	27	0	0	0	0	0	0	0	0
1	9	81	729	1	4	16	64	0	0	0	0	0	0	0	0
1	10	100	1000	1	5	25	125	1	0	0	0	0	0	0	0
1	11	121	1331	1	6	36	216	1	1	1	1	0	0	0	0
1	12	144	1728	1	7	49	343	1	2	4	8	0	0	0	0
1	13	169	2197	1	8	64	512	1	3	9	27	0	0	0	0
1	14	196	2744	1	9	81	729	1	4	16	64	0	0	0	0
1	15	225	3375	1	10	100	1000	1	5	25	125	1	0	0	0
1	16	256	4096	1	11	121	1331	1	6	36	216	1	1	1	1
1	17	289	4913	1	12	144	1728	1	7	49	343	1	2	4	8
1	18	324	5832	1	13	169	2197	1	8	64	512	1	3	9	27
1	19	361	6859	1	14	196	2744	1	9	81	729	1	4	16	64
1	20	400	8000	1	15	225	3375	1	10	100	1000	1	5	25	125

Speaker notes

This is what the raw numbers look like. I apologize for the size of the fonts. I want you to focus just on the patterns shown here.

06-02 What the numbers look like, overall terms

Overall Intercept	Overall Slope	Overall Quadratic	Overall Cubic	Knot 1 Intercept	Knot 1 Slope	Knot 1 Quadratic	Knot 1 Cubic	Knot 2 Intercept	Knot 2 Slope	Knot 2 Quadratic	Knot 2 Cubic	Knot 3 Intercept	Knot 3 Slope	Knot 3 Quadratic	Knot 3 Cubic
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	2	4	8	0	0	0	0	0	0	0	0	0	0	0	0
1	3	9	27	0	0	0	0	0	0	0	0	0	0	0	0
1	4	16	64	0	0	0	0	0	0	0	0	0	0	0	0
1	5	25	125	1	0	0	0	0	0	0	0	0	0	0	0
1	6	36	216	1	1	1	1	0	0	0	0	0	0	0	0
1	7	49	343	1	2	4	8	0	0	0	0	0	0	0	0
1	8	64	512	1	3	9	27	0	0	0	0	0	0	0	0
1	9	81	729	1	4	16	64	0	0	0	0	0	0	0	0
1	10	100	1000	1	5	25	125	1	0	0	0	0	0	0	0
1	11	121	1331	1	6	36	216	1	1	1	1	0	0	0	0
1	12	144	1728	1	7	49	343	1	2	4	8	0	0	0	0
1	13	169	2197	1	8	64	512	1	3	9	27	0	0	0	0
1	14	196	2744	1	9	81	729	1	4	16	64	0	0	0	0
1	15	225	3375	1	10	100	1000	1	5	25	125	1	0	0	0
1	16	256	4096	1	11	121	1331	1	6	36	216	1	1	1	1
1	17	289	4913	1	12	144	1728	1	7	49	343	1	2	4	8
1	18	324	5832	1	13	169	2197	1	8	64	512	1	3	9	27
1	19	361	6859	1	14	196	2744	1	9	81	729	1	4	16	64
1	20	400	8000	1	15	225	3375	1	10	100	1000	1	5	25	125

Speaker notes

The overall terms are constant, linear, quadratic, or cubic across the entire range of the data.

06-02 What the numbers look like, knot 1 terms

Overall Intercept	Overall Slope	Overall Quadratic	Overall Cubic	Knot 1 Intercept	Knot 1 Slope	Knot 1 Quadratic	Knot 1 Cubic	Knot 2 Intercept	Knot 2 Slope	Knot 2 Quadratic	Knot 2 Cubic	Knot 3 Intercept	Knot 3 Slope	Knot 3 Quadratic	Knot 3 Cubic
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	2	4	8	0	0	0	0	0	0	0	0	0	0	0	0
1	3	9	27	0	0	0	0	0	0	0	0	0	0	0	0
1	4	16	64	0	0	0	0	0	0	0	0	0	0	0	0
1	5	25	125	1	0	0	0	0	0	0	0	0	0	0	0
1	6	36	216	1	1	1	1	0	0	0	0	0	0	0	0
1	7	49	343	1	2	4	8	0	0	0	0	0	0	0	0
1	8	64	512	1	3	9	27	0	0	0	0	0	0	0	0
1	9	81	729	1	4	16	64	0	0	0	0	0	0	0	0
1	10	100	1000	1	5	25	125	1	0	0	0	0	0	0	0
1	11	121	1331	1	6	36	216	1	1	1	1	0	0	0	0
1	12	144	1728	1	7	49	343	1	2	4	8	0	0	0	0
1	13	169	2197	1	8	64	512	1	3	9	27	0	0	0	0
1	14	196	2744	1	9	81	729	1	4	16	64	0	0	0	0
1	15	225	3375	1	10	100	1000	1	5	25	125	1	0	0	0
1	16	256	4096	1	11	121	1331	1	6	36	216	1	1	1	1
1	17	289	4913	1	12	144	1728	1	7	49	343	1	2	4	8
1	18	324	5832	1	13	169	2197	1	8	64	512	1	3	9	27
1	19	361	6859	1	14	196	2744	1	9	81	729	1	4	16	64
1	20	400	8000	1	15	225	3375	1	10	100	1000	1	5	25	125

Speaker notes

The knot 1 terms are all zero until you hit x=5. Then they are constant, linear, quadratic, or cubic across the rest of the data.

06-02 What the numbers look like, knot 2 terms

Overall Intercept	Overall Slope	Overall Quadratic	Overall Cubic	Knot 1 Intercept	Knot 1 Slope	Knot 1 Quadratic	Knot 1 Cubic	Knot 2 Intercept	Knot 2 Slope	Knot 2 Quadratic	Knot 2 Cubic	Knot 3 Intercept	Knot 3 Slope	Knot 3 Quadratic	Knot 3 Cubic
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	2	4	8	0	0	0	0	0	0	0	0	0	0	0	0
1	3	9	27	0	0	0	0	0	0	0	0	0	0	0	0
1	4	16	64	0	0	0	0	0	0	0	0	0	0	0	0
1	5	25	125	1	0	0	0	0	0	0	0	0	0	0	0
1	6	36	216	1	1	1	1	0	0	0	0	0	0	0	0
1	7	49	343	1	2	4	8	0	0	0	0	0	0	0	0
1	8	64	512	1	3	9	27	0	0	0	0	0	0	0	0
1	9	81	729	1	4	16	64	0	0	0	0	0	0	0	0
1	10	100	1000	1	5	25	125	1	0	0	0	0	0	0	0
1	11	121	1331	1	6	36	216	1	1	1	1	0	0	0	0
1	12	144	1728	1	7	49	343	1	2	4	8	0	0	0	0
1	13	169	2197	1	8	64	512	1	3	9	27	0	0	0	0
1	14	196	2744	1	9	81	729	1	4	16	64	0	0	0	0
1	15	225	3375	1	10	100	1000	1	5	25	125	1	0	0	0
1	16	256	4096	1	11	121	1331	1	6	36	216	1	1	1	1
1	17	289	4913	1	12	144	1728	1	7	49	343	1	2	4	8
1	18	324	5832	1	13	169	2197	1	8	64	512	1	3	9	27
1	19	361	6859	1	14	196	2744	1	9	81	729	1	4	16	64
1	20	400	8000	1	15	225	3375	1	10	100	1000	1	5	25	125

Speaker notes

The knot 1 terms are all zero until you hit x=10. Then they are constant, linear, quadratic, or cubic across the rest of the data.

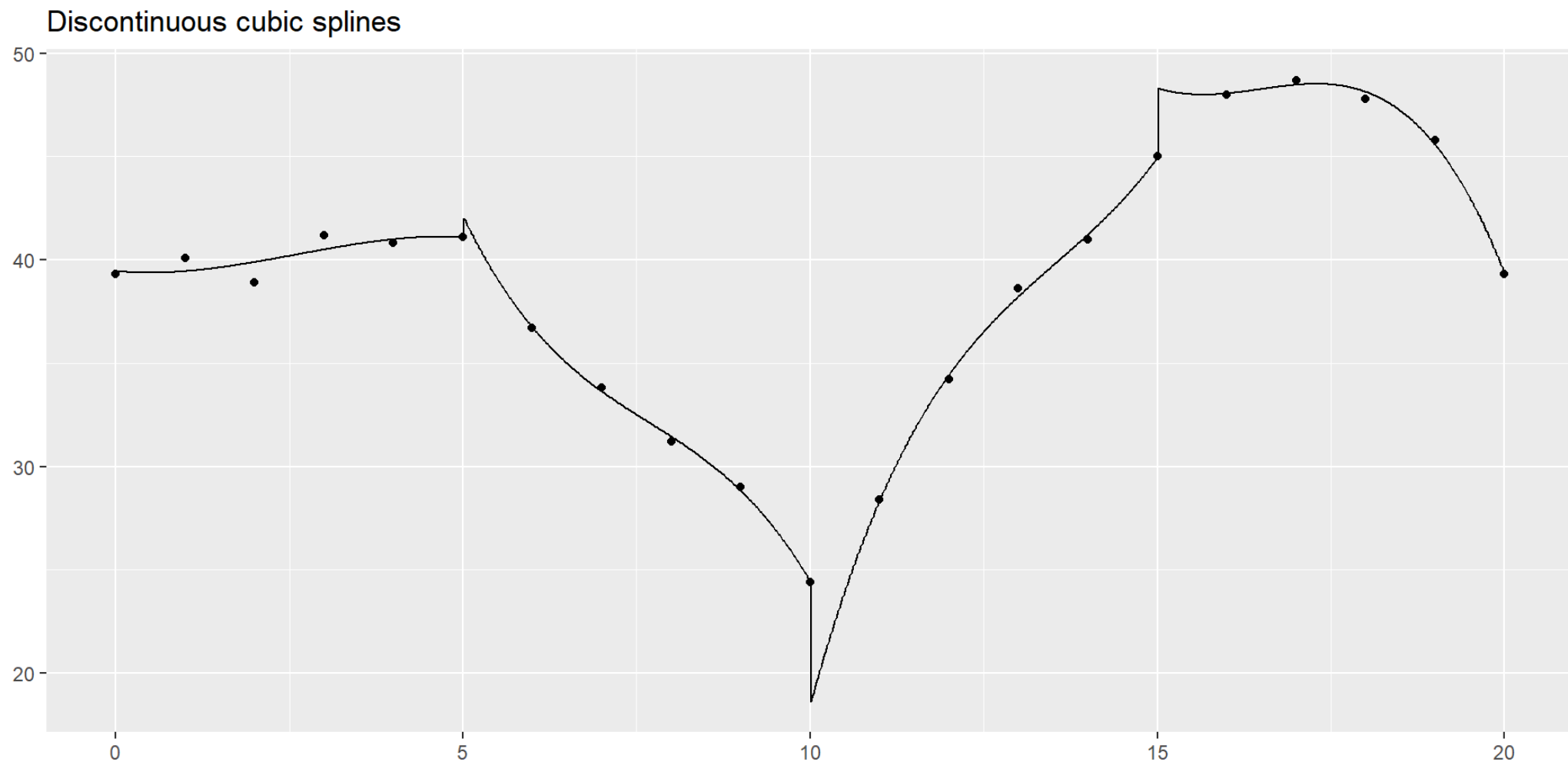
06-02 What the numbers look like, knot 3 terms

Overall Intercept	Overall Slope	Overall Quadratic	Overall Cubic	Knot 1 Intercept	Knot 1 Slope	Knot 1 Quadratic	Knot 1 Cubic	Knot 2 Intercept	Knot 2 Slope	Knot 2 Quadratic	Knot 2 Cubic	Knot 3 Intercept	Knot 3 Slope	Knot 3 Quadratic	Knot 3 Cubic
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	2	4	8	0	0	0	0	0	0	0	0	0	0	0	0
1	3	9	27	0	0	0	0	0	0	0	0	0	0	0	0
1	4	16	64	0	0	0	0	0	0	0	0	0	0	0	0
1	5	25	125	1	0	0	0	0	0	0	0	0	0	0	0
1	6	36	216	1	1	1	1	0	0	0	0	0	0	0	0
1	7	49	343	1	2	4	8	0	0	0	0	0	0	0	0
1	8	64	512	1	3	9	27	0	0	0	0	0	0	0	0
1	9	81	729	1	4	16	64	0	0	0	0	0	0	0	0
1	10	100	1000	1	5	25	125	1	0	0	0	0	0	0	0
1	11	121	1331	1	6	36	216	1	1	1	1	0	0	0	0
1	12	144	1728	1	7	49	343	1	2	4	8	0	0	0	0
1	13	169	2197	1	8	64	512	1	3	9	27	0	0	0	0
1	14	196	2744	1	9	81	729	1	4	16	64	0	0	0	0
1	15	225	3375	1	10	100	1000	1	5	25	125	1	0	0	0
1	16	256	4096	1	11	121	1331	1	6	36	216	1	1	1	1
1	17	289	4913	1	12	144	1728	1	7	49	343	1	2	4	8
1	18	324	5832	1	13	169	2197	1	8	64	512	1	3	9	27
1	19	361	6859	1	14	196	2744	1	9	81	729	1	4	16	64
1	20	400	8000	1	15	225	3375	1	10	100	1000	1	5	25	125

Speaker notes

The knot 3 terms are all zero until you hit x=15. Then they are constant, linear, quadratic, or cubic across the rest of the data.

06-02 Discontinuous cubic spline



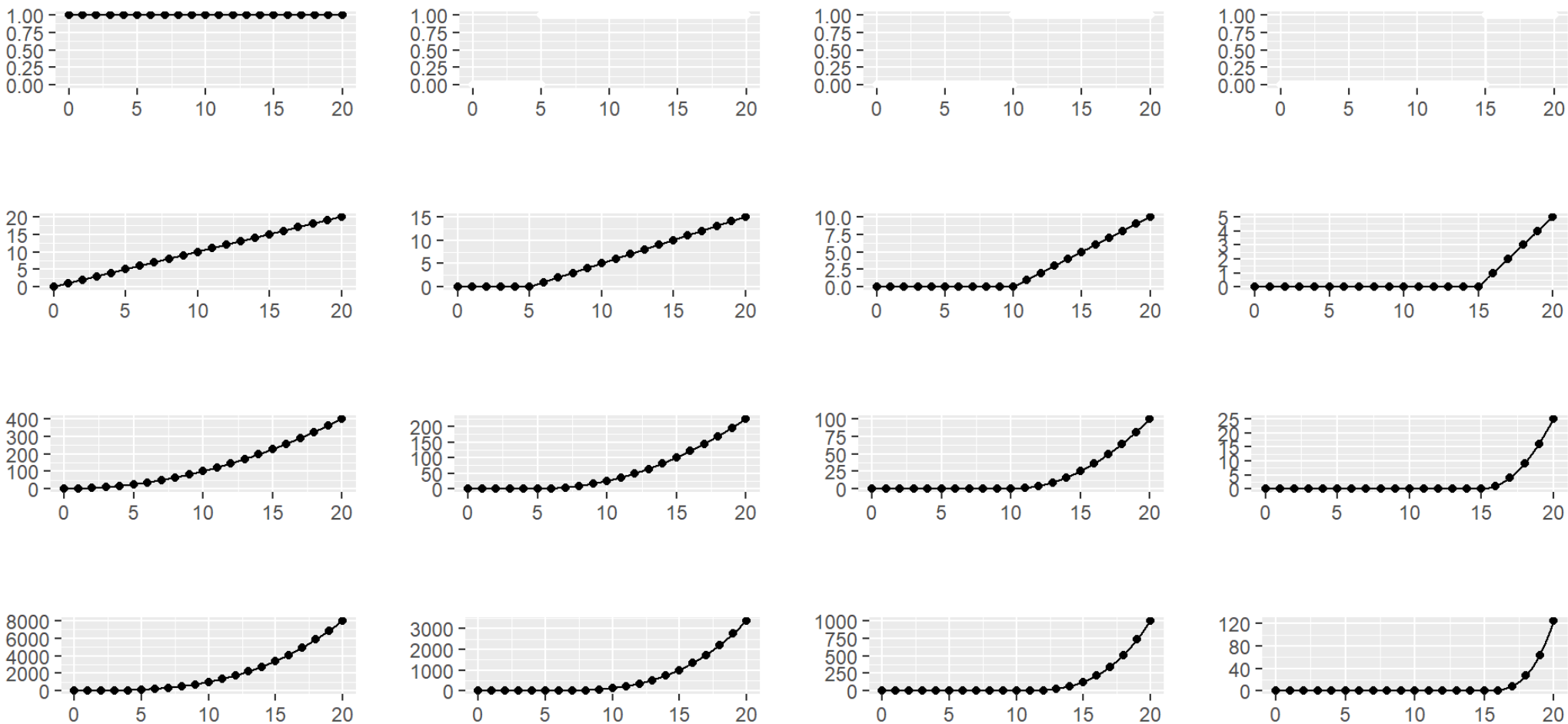
Speaker notes

So, we could fit a cubic model for the first five data points, for the second five, the third five, and the fourth five. This is a bit much: a cubic model has four parameters, so fitting four of them would use up 16 degrees of freedom in a data set with only 20 observations. But bear with me a bit on this.

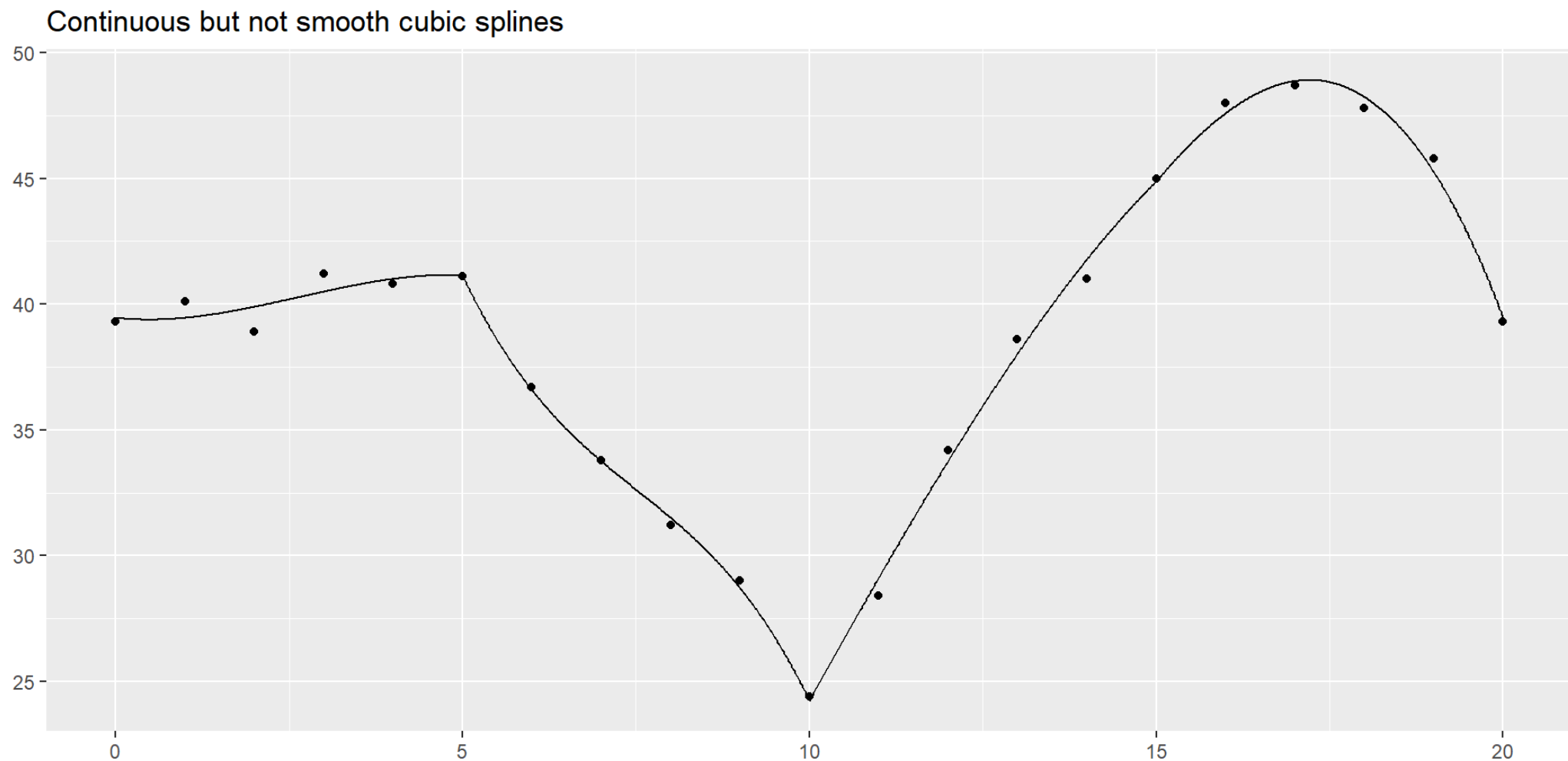
The trick to fitting four separate cubic polynomials is to “restart” the intercept, linear, quadratic, and cubic terms after x=5, x=10, and x=15, as shown above. This leads to a model with 16 degrees of freedom. This is way too many degrees of freedom for only 20 data points, but it helps anchor a series of more reasonable models.

This function is not continuous or smooth. To make the function continuous, drop the extra intercept terms.

06-02 Remove piece-wise intercepts



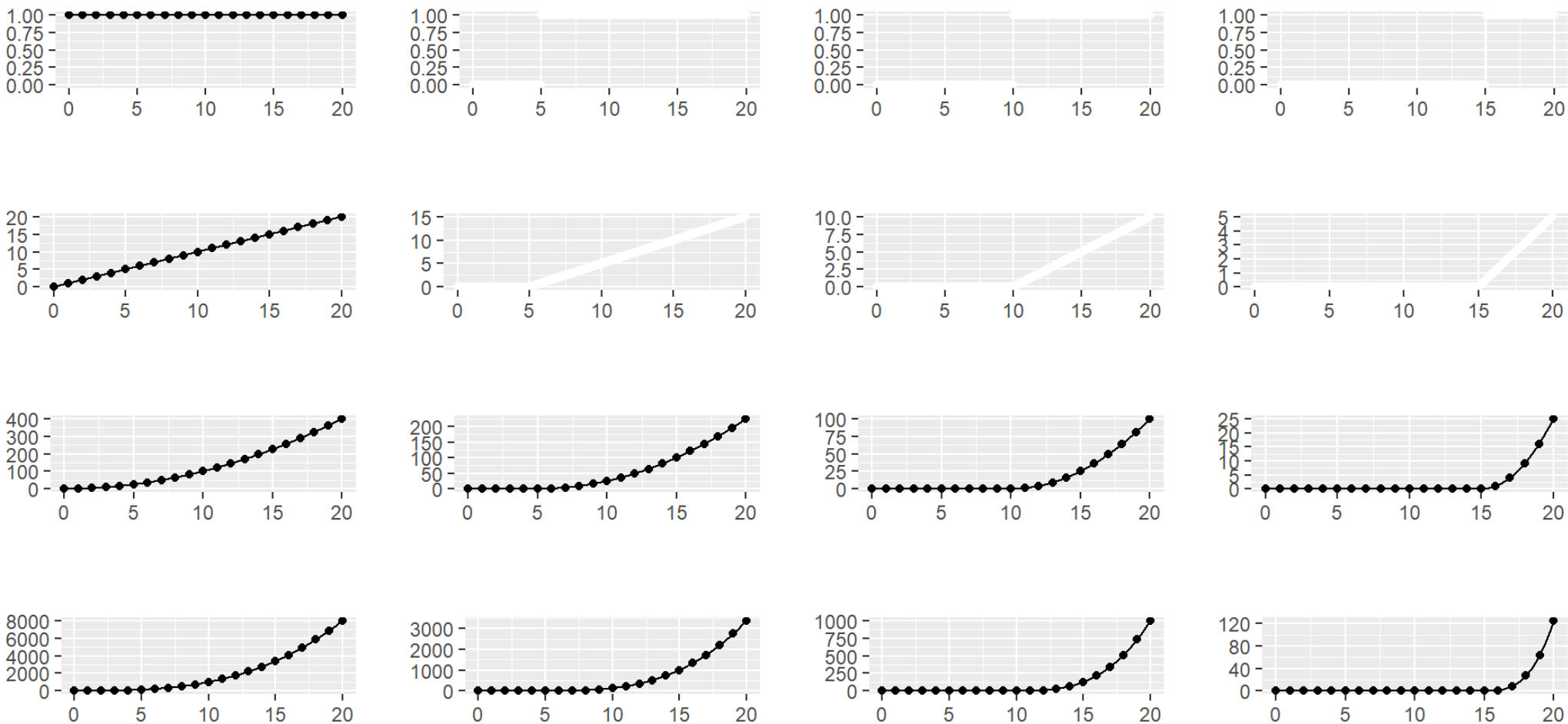
06-02 Continuous spline



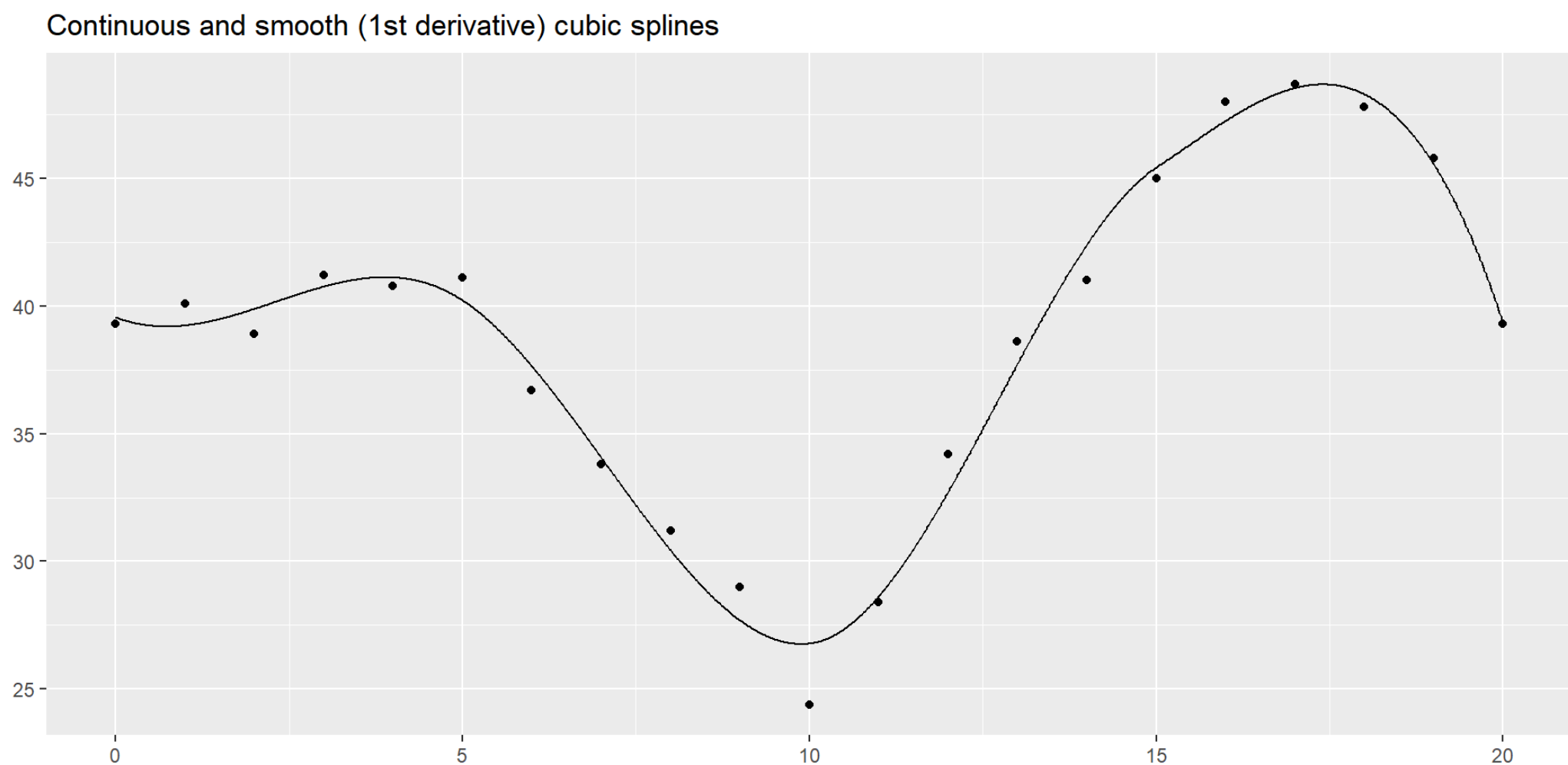
Speaker notes

While this graph is continuous, it still takes some abrupt turns. What this curve lacks is smoothness. The mathematical concept of smoothness is measured in terms of the continuity of derivatives.

06-02 Remove piecewise linear terms



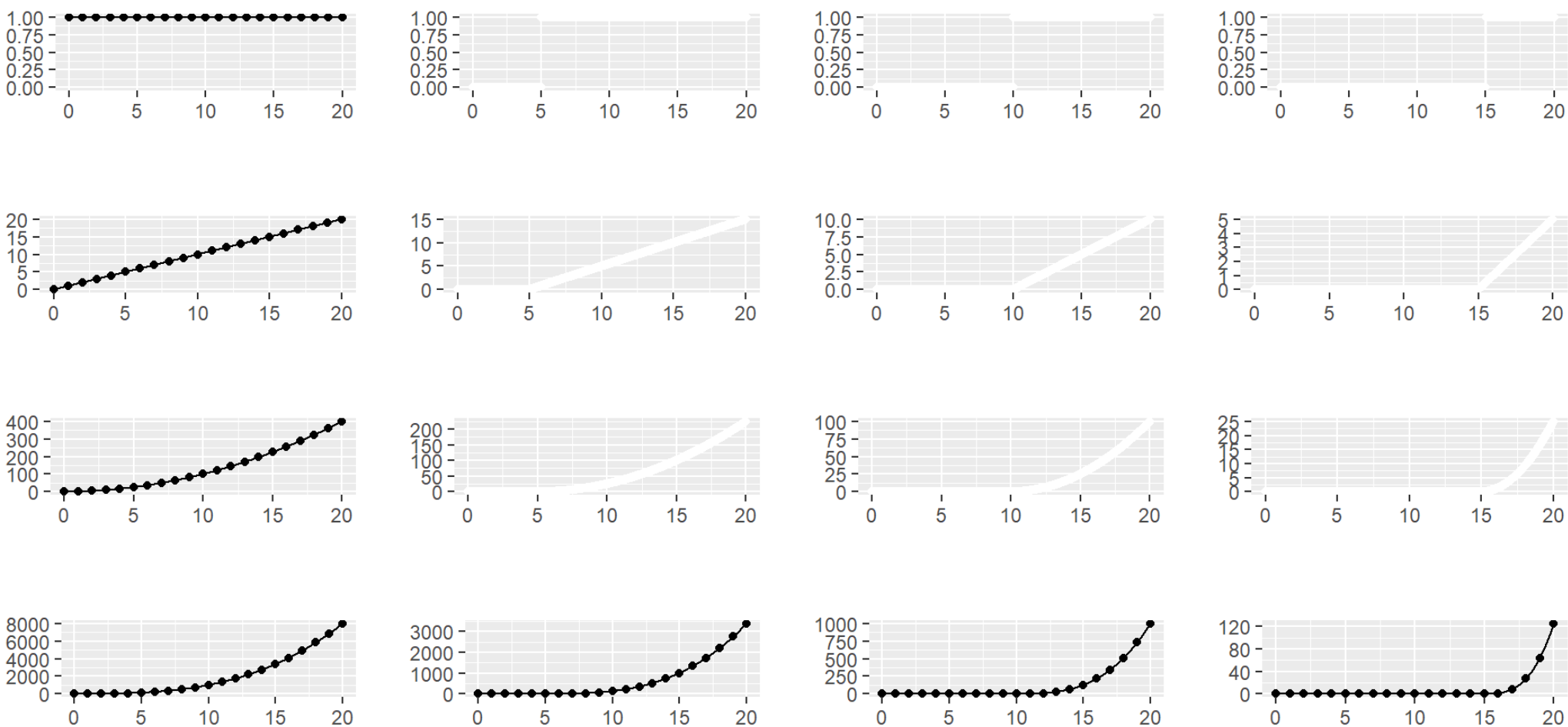
06-02 Smooth splines



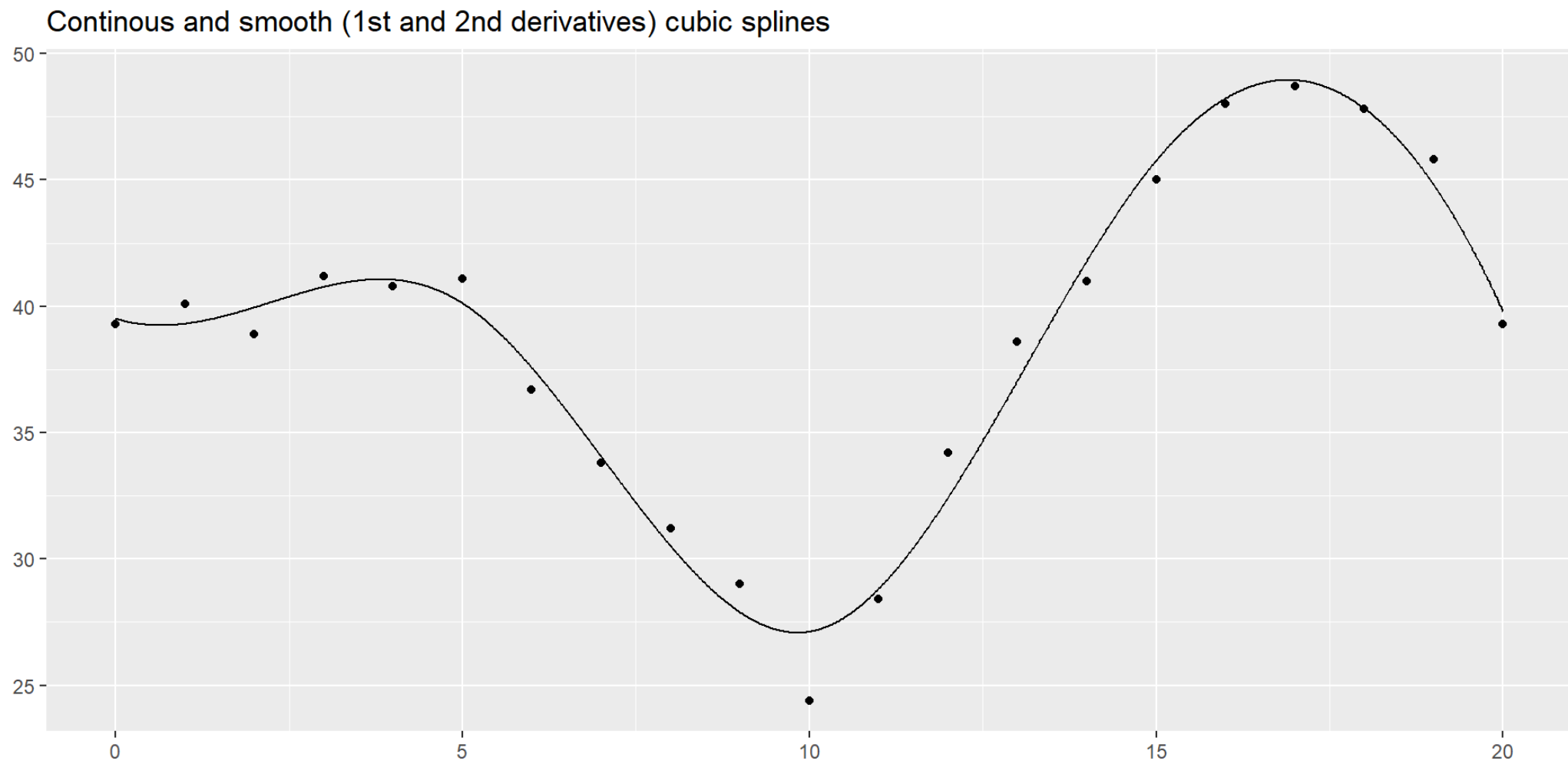
Speaker notes

Here is a function that has a continuous first derivative. You fit this model by dropping the extra linear terms beyond the first one. Notice a pattern here. As you place additional restrictions on the spline (continuity, smoothness), you need fewer parameters. The four cubic models with no restrictions used up 16 degrees of freedom. When you added a continuity restriction, you only needed 13 degrees of freedom for the model. Add a smoothness restriction and you only need 10 degrees of freedom.

06-02 Remove piecewise quadratic terms



06-02 Even smoother



Speaker notes

Here is a function that has continuous first and second derivatives. This is a greater degree of smoothness than above and it requires only 7 degrees of freedom.

This is what most people refer to when they talk about splines: a piecewise cubic model with continuity and continuous first and second derivatives. It is a fairly simple model (not that many degrees of freedom), but it produces a curve that has the flexibility to fit a variety of curves that have the aesthetically pleasing features of continuity and smoothness.

Continuity and smoothness are more than just aesthetics, though. There are many scientific settings where we expect no jumps (discontinuities) and no abrupt turns (lack of smoothness). If you are measuring the onset of symptoms from a disease, you know that the viruses or bacteria that are causing the disease are increasing in a continuous and smooth pattern. So any problems that they cause should also increase in a continuous and smooth pattern.

Other settings, however, should not necessarily be expected to produce continuous and smooth outcomes. If a particular metabolic pathway becomes saturated or an anotomical barrier is breached, the suddenness transition could result in an abrupt turn or a discontinuity. So do think about the particular context of your problem when deciding what type of spline model to use.

This approach is simple and easy to follow, but there is one catch. There is an issue with multicollinearity.

06-02 Multicollinearity

```
     S1   S2   S3   S4   S5   S6
S1 1.00 0.97 0.92 0.98 0.73 0.55
S2 0.97 1.00 0.99 0.99 0.86 0.67
S3 0.92 0.99 1.00 0.96 0.93 0.76
S4 0.98 0.99 0.96 1.00 0.79 0.59
S5 0.73 0.86 0.93 0.79 1.00 0.92
S6 0.55 0.67 0.76 0.59 0.92 1.00
```

Speaker notes

The correlations are quite high and this can lead to computational problems, including rounding errors. So most spline models implemented on a computer use a different approach.

Break #2

- What you have learned
  - Constructing a spline from small pieces
- What’s coming next
  - B-splines

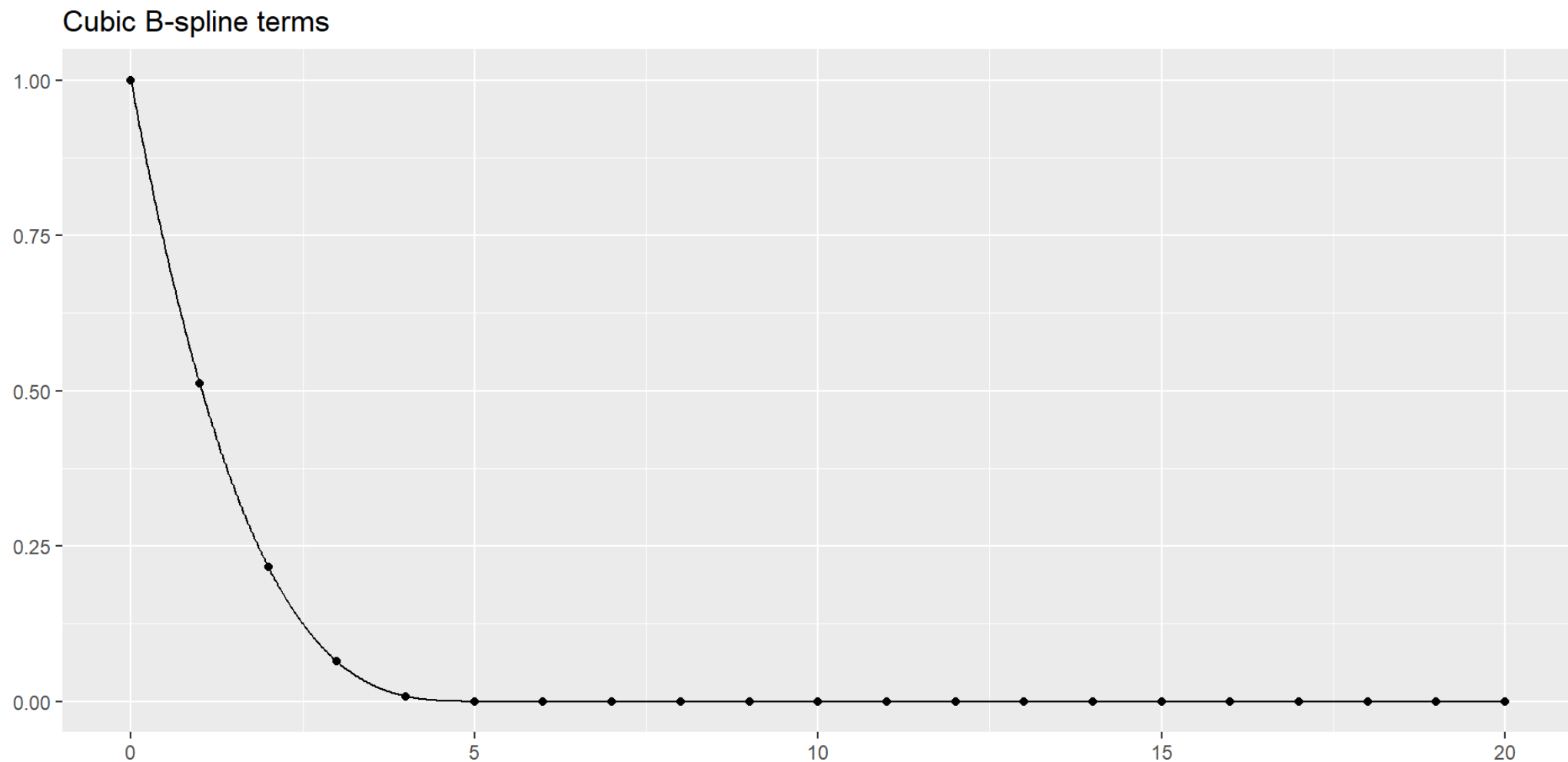
06-03 B-splines

- B-splines = Basis splines
  - Basis is a mathematical property of a set and a subset
- Mathematically equivalent to formulation shown earlier
- Much less collinearity

Speaker notes

B-splines provide a solution with less issues of multi-collinearity.

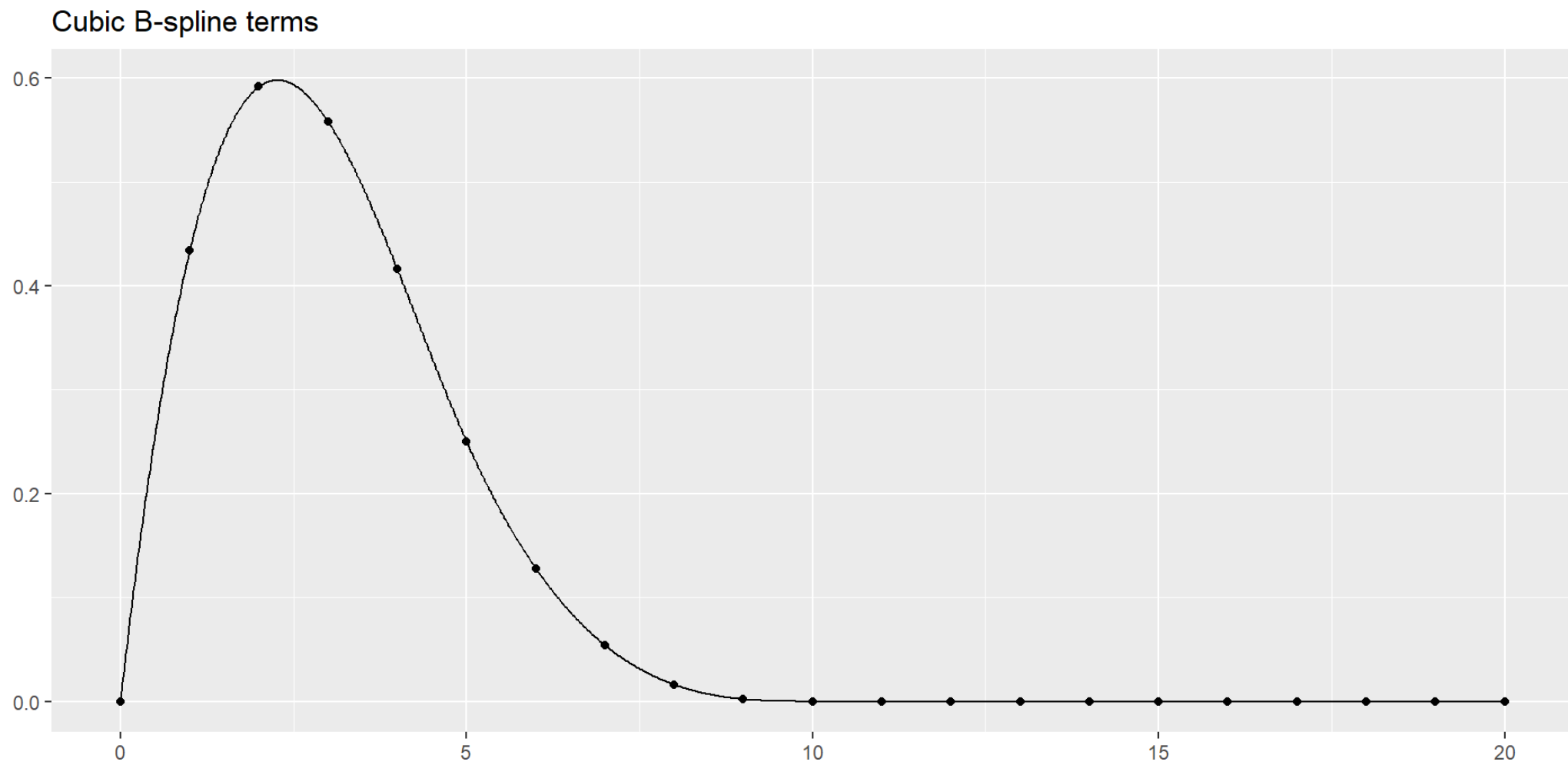
06-03 B-spline, 1



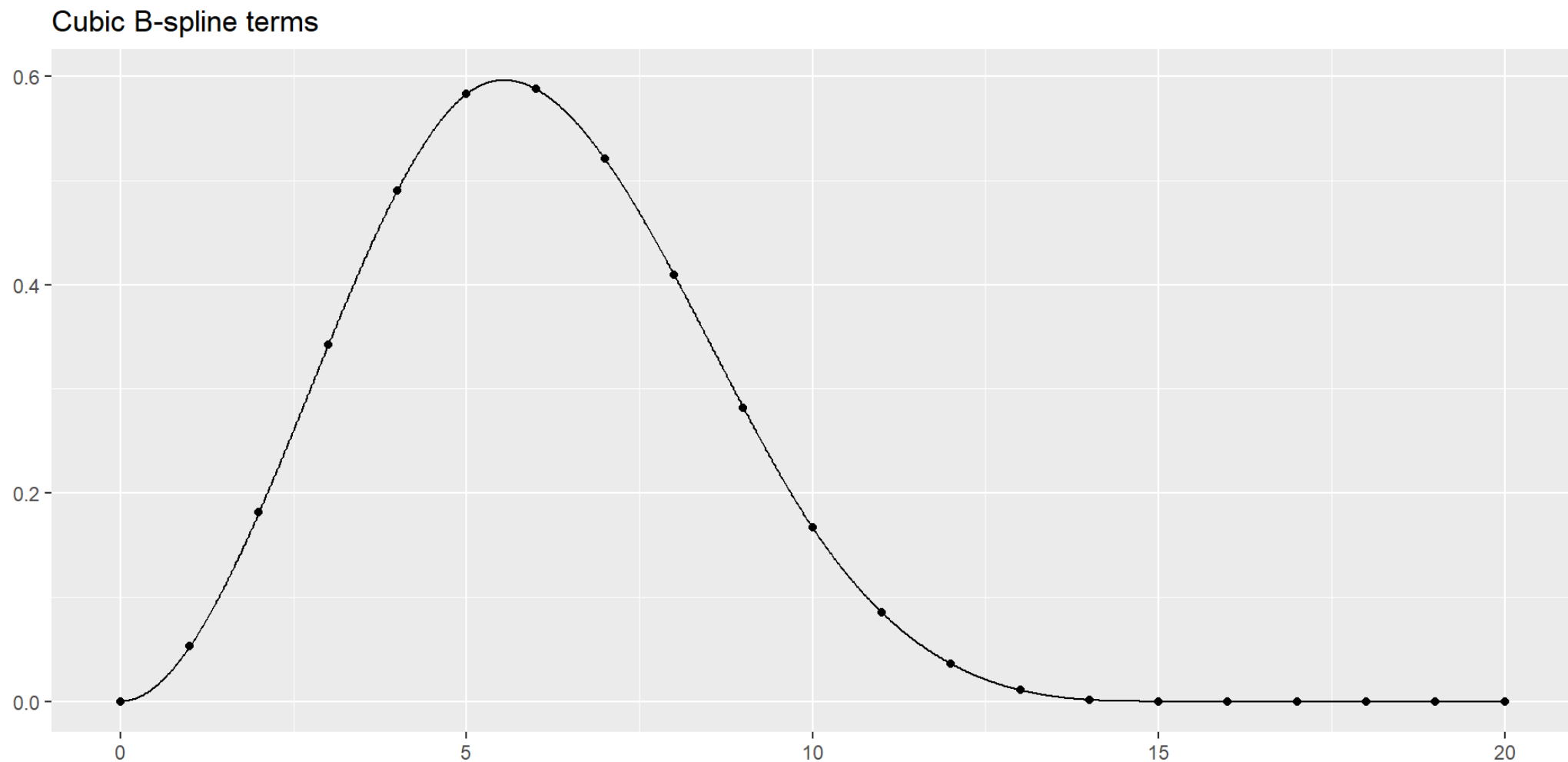
Speaker notes

The individual columns represent piecwise cubic polynomials. As I cycle through these graphs, notice how they are concentrated in certain intervals and there is only a partial overlap between these intervals. Also notice how they transition smoothly to zero outside those intervals.

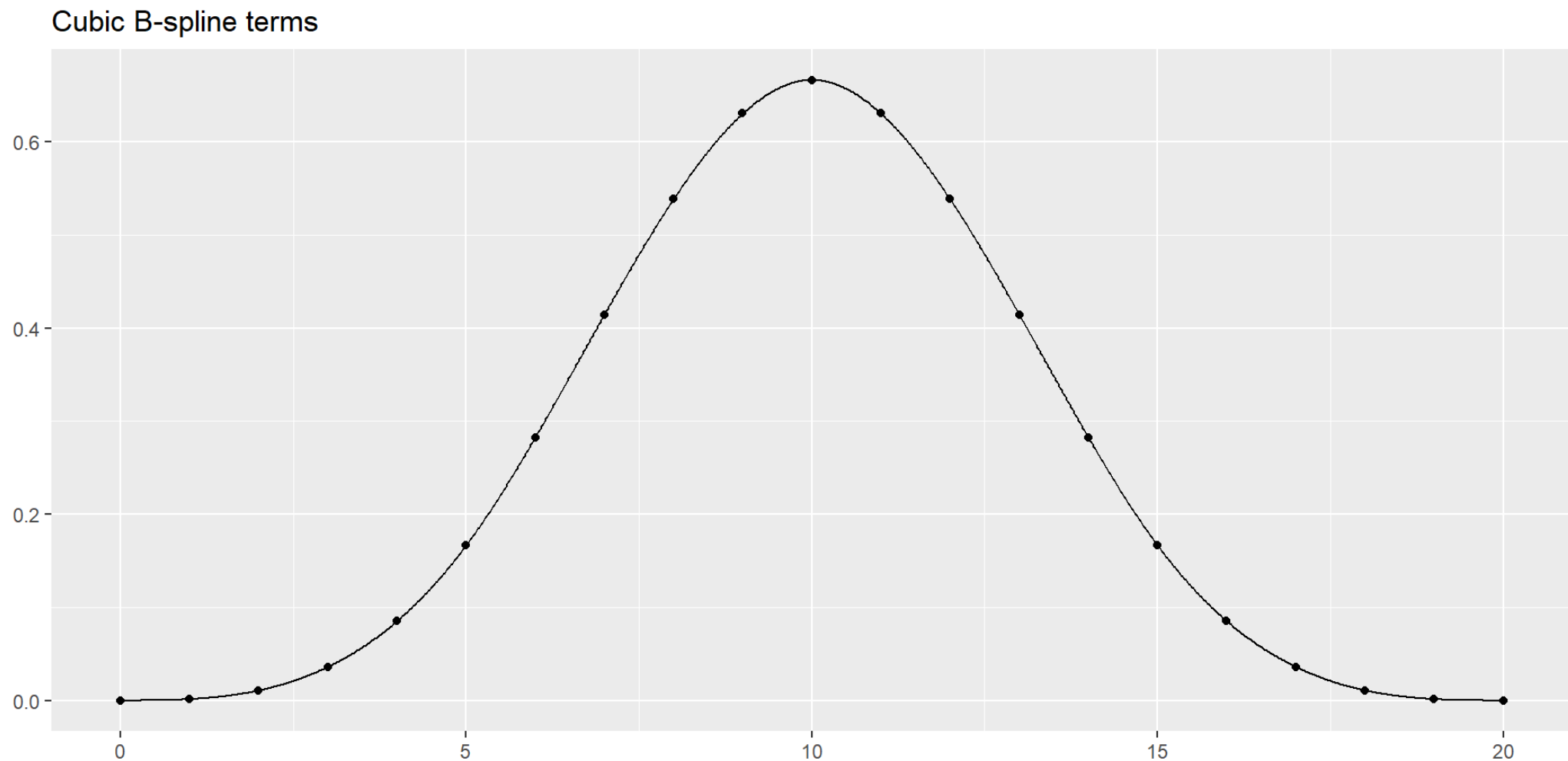
06-03 B-spline, 2



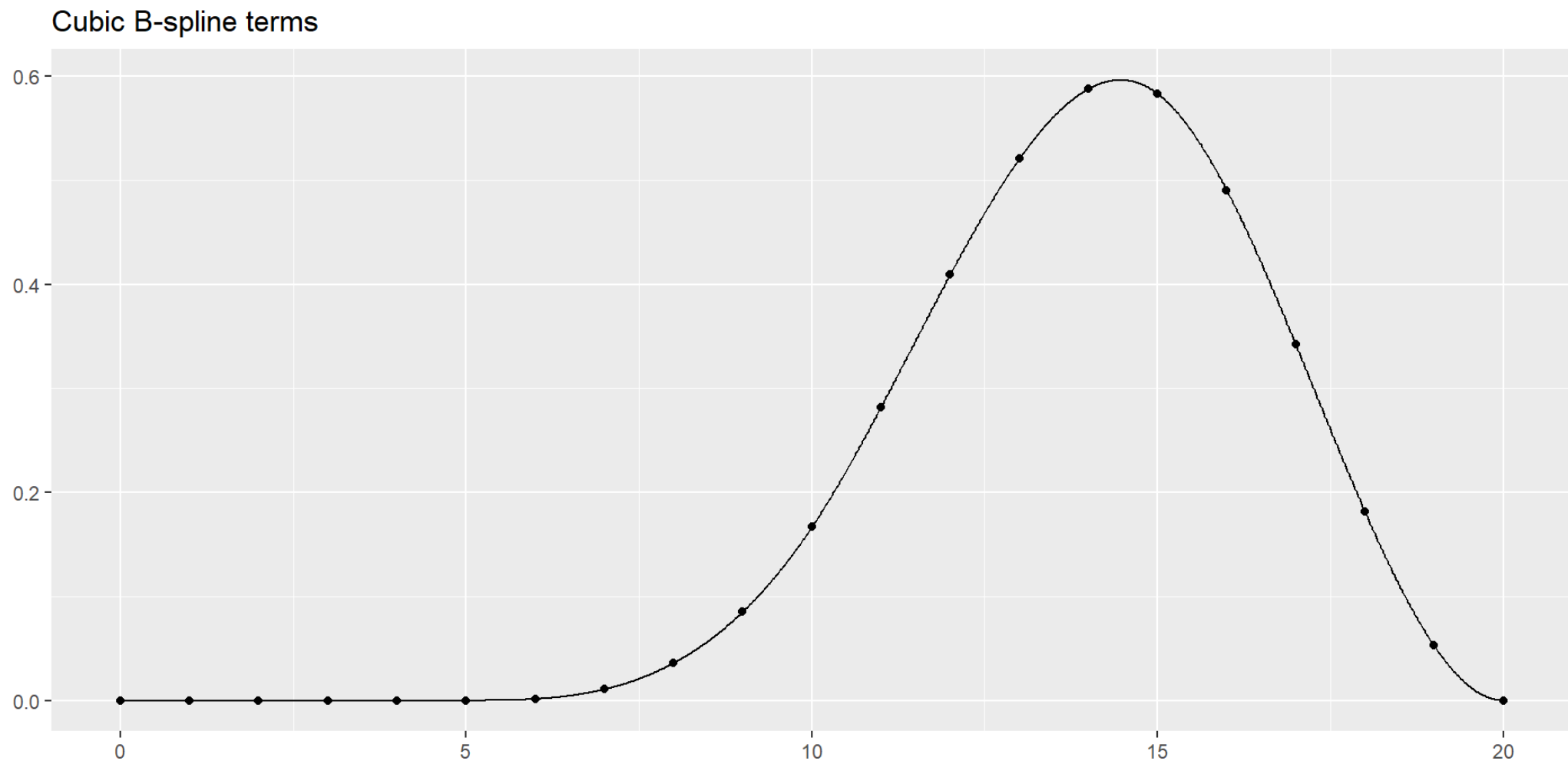
06-03 B-spline, 3



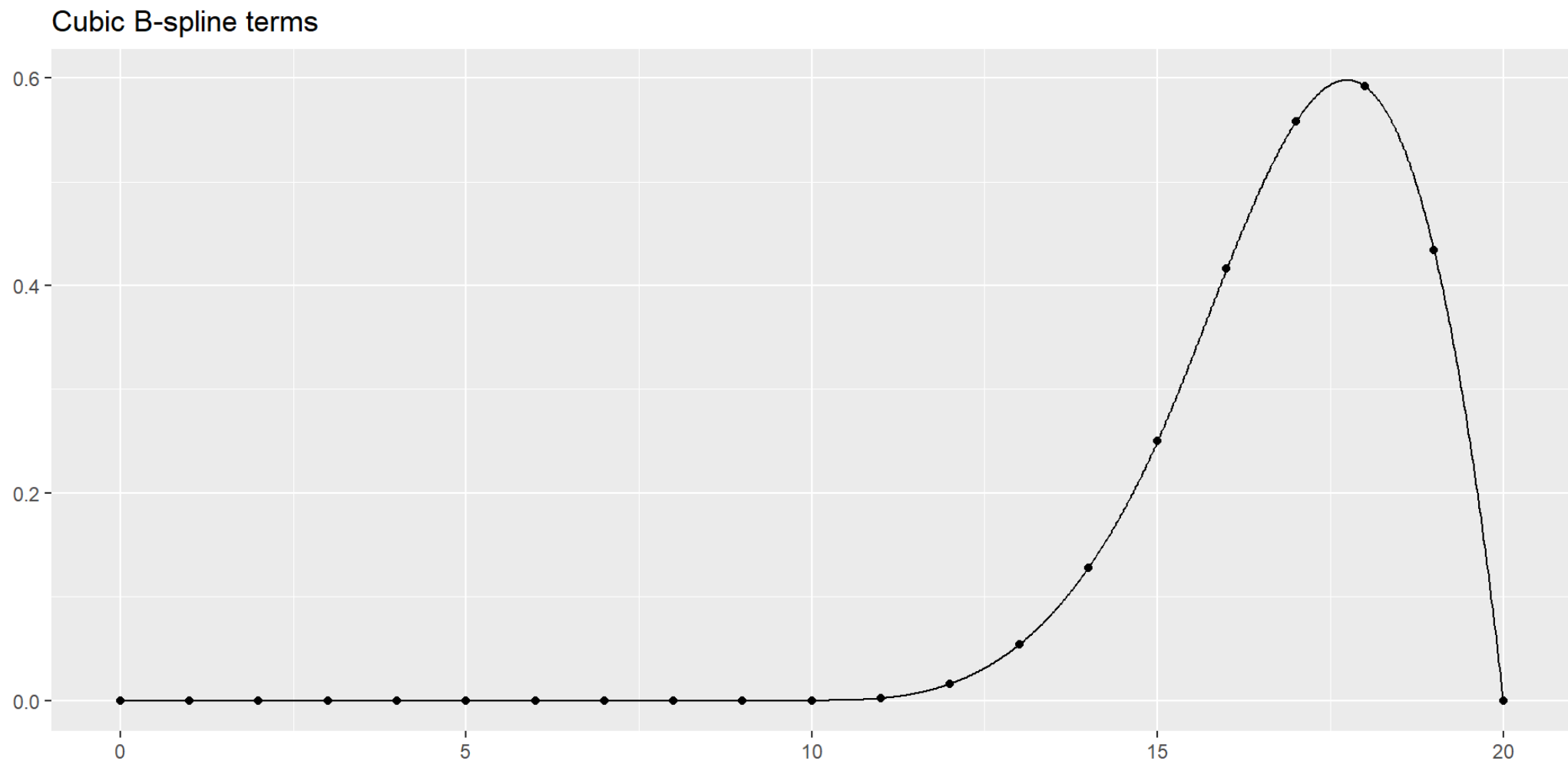
06-03 B-spline, 4



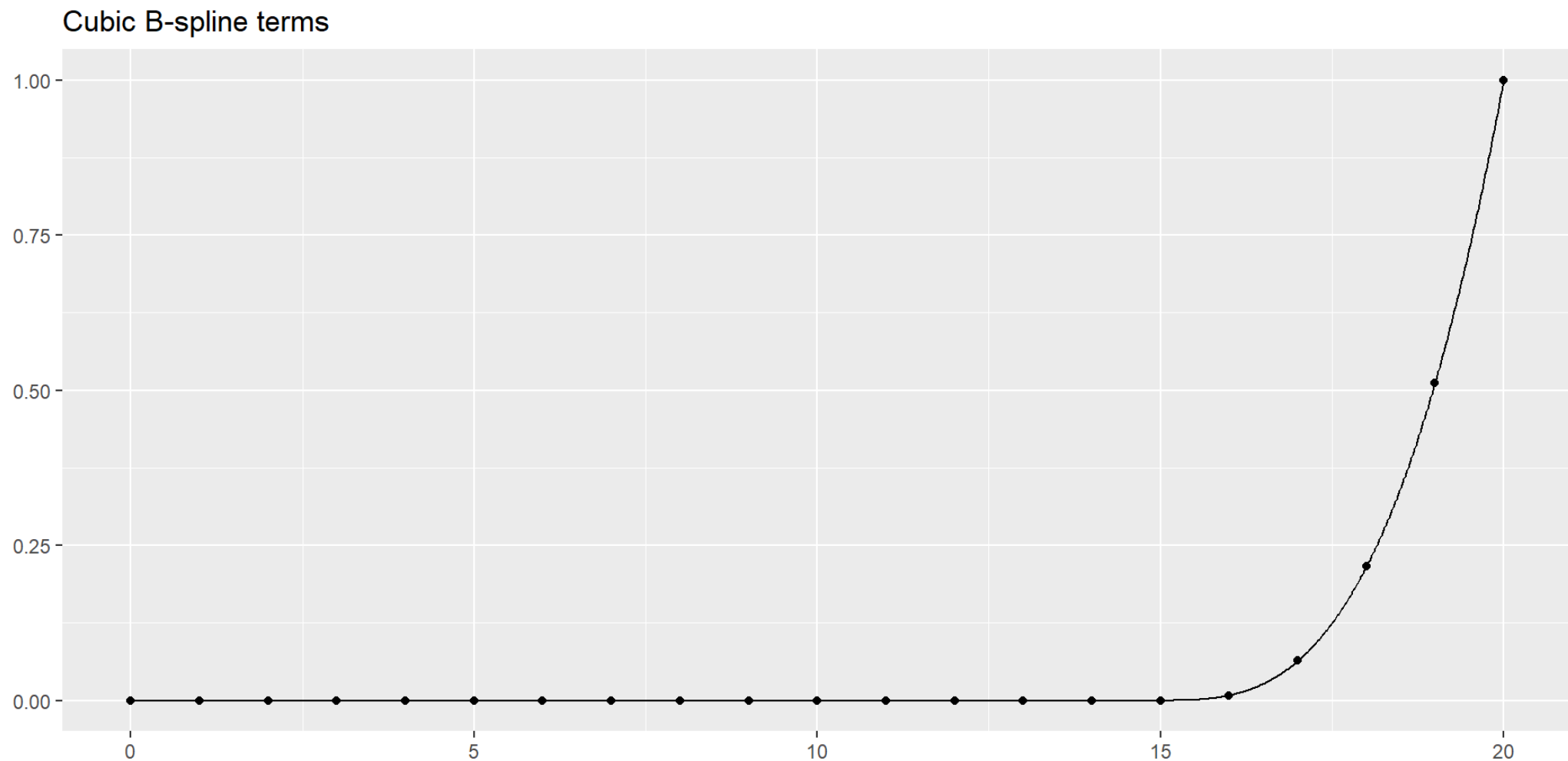
06-03 B-spline, 5



06-03 B-spline, 6



06-03 B-spline, 7



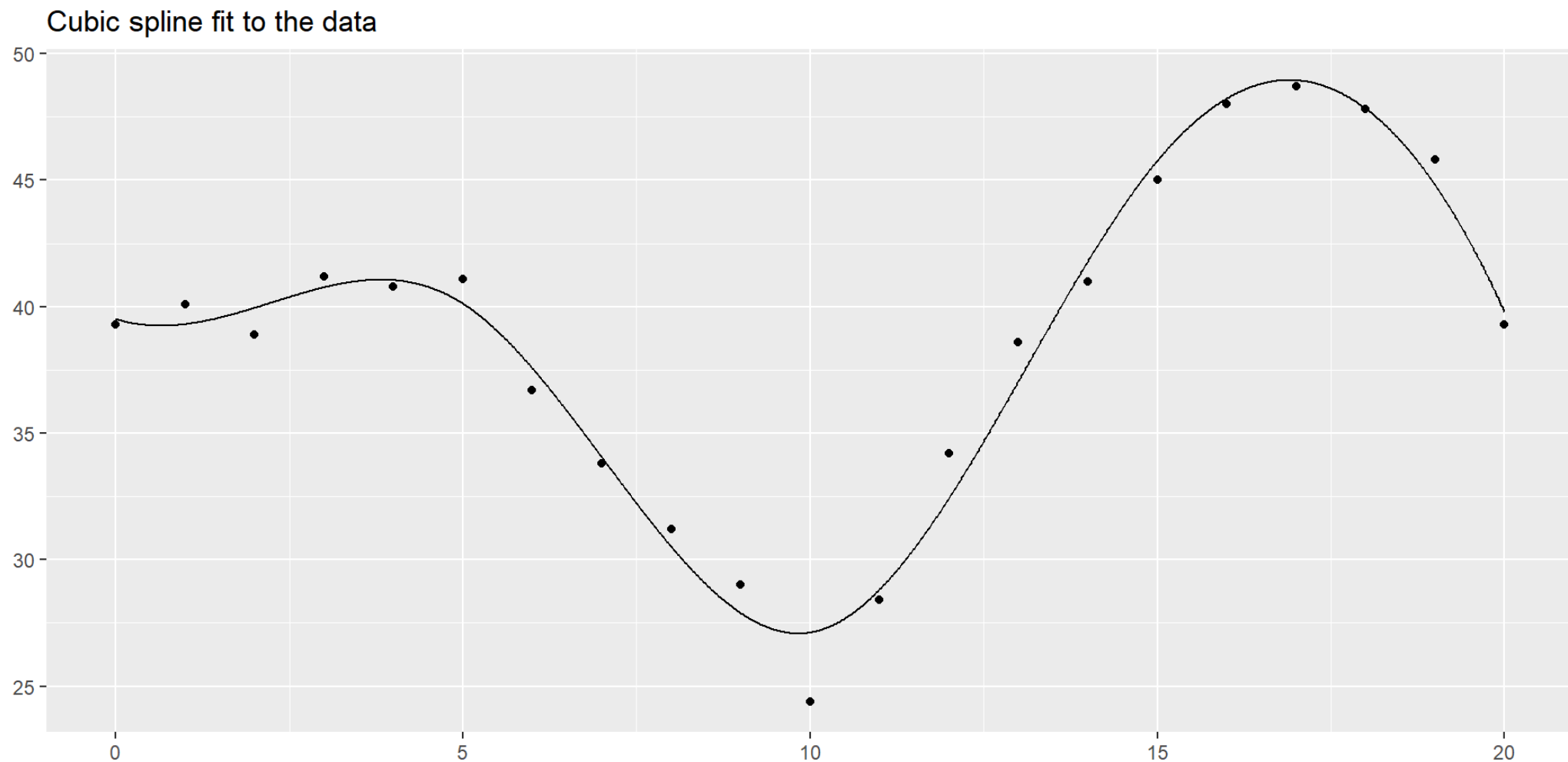
06-03 Correlation of B-spline terms

	S1	S2	S3	S4	S5	S6	S7
S1	1.00	0.39	-0.19	-0.36	-0.30	-0.22	-0.12
S2	0.39	1.00	0.36	-0.47	-0.54	-0.39	-0.22
S3	-0.19	0.36	1.00	0.18	-0.63	-0.54	-0.30
S4	-0.36	-0.47	0.18	1.00	0.18	-0.47	-0.36
S5	-0.30	-0.54	-0.63	0.18	1.00	0.36	-0.19
S6	-0.22	-0.39	-0.54	-0.47	0.36	1.00	0.39
S7	-0.12	-0.22	-0.30	-0.36	-0.19	0.39	1.00

Speaker notes

Although there is some correlation, this is not nearly as bad as the piecewise approach.

06-03 B-spline fit to the artificial data



Break #3

- What you have learned
  - B-splines
- What’s coming next
  - Placement and number of knots

06-04 How many knots?

- Use AIC or BIC
- Eyeball the data
- Frank Harrell's suggestion
  - Use 4 knots if $n < 100$
  - Otherwise use 5 knots
- Use your a priori beliefs
- Looking at some preliminary graphs

Speaker notes

A difficult question is how many knots to use. Too many knots and you might end up overfitting. Too few and you might end up with not enough flexibility to fit your data well.

The AIC (Akaike Information Criterion) and the BIC (Bayesian Information Criterion) are useful measures for comparing different statistical models. In linear regression both AIC and BIC look at how close the fitted curve is to the data, but adds a penalty for model complexity. This helps avoid the situation where an excessively complex model with only marginally better predictive power is selected over a simpler model that predicts almost as well.

Note that a p-value will not work here except in some special cases where all of the knots but one coincide. The p-value fails because (with a few rare exceptions) one spline model is not nested inside another.

Look at the number of bends in the data. If the data increases to a single maximum and then decreases after that, a simpler spline with 2 or 3 knots may be sufficient. This also applies if the data decreases to a single minimum and then increases after that. If there are more bends (e.g., increase to a maximum, decrease to a minimum, and then increase again), then a larger number of knots may be needed.

Frank Harrell has a simple suggestion. Use 4 knots for small data sets. Small means $n < 100$. Use 5 knots for large data sets.

You may have a feel for how much complexity is appropriate based on your years of experience as a data analyst and your scientific knowledge of the process at hand. After you work with enough splines, you do get an appreciation on how wiggly they can get. If you also have a rough idea of how the nonlinear relationship is going to be, perhaps based on seeing other similar problems in the area, you can match the degrees of freedom of the spline to your expectation, prior to looking at the data.

Sometimes you have knowledge of the specific application that will help you to figure out where to put your knots.

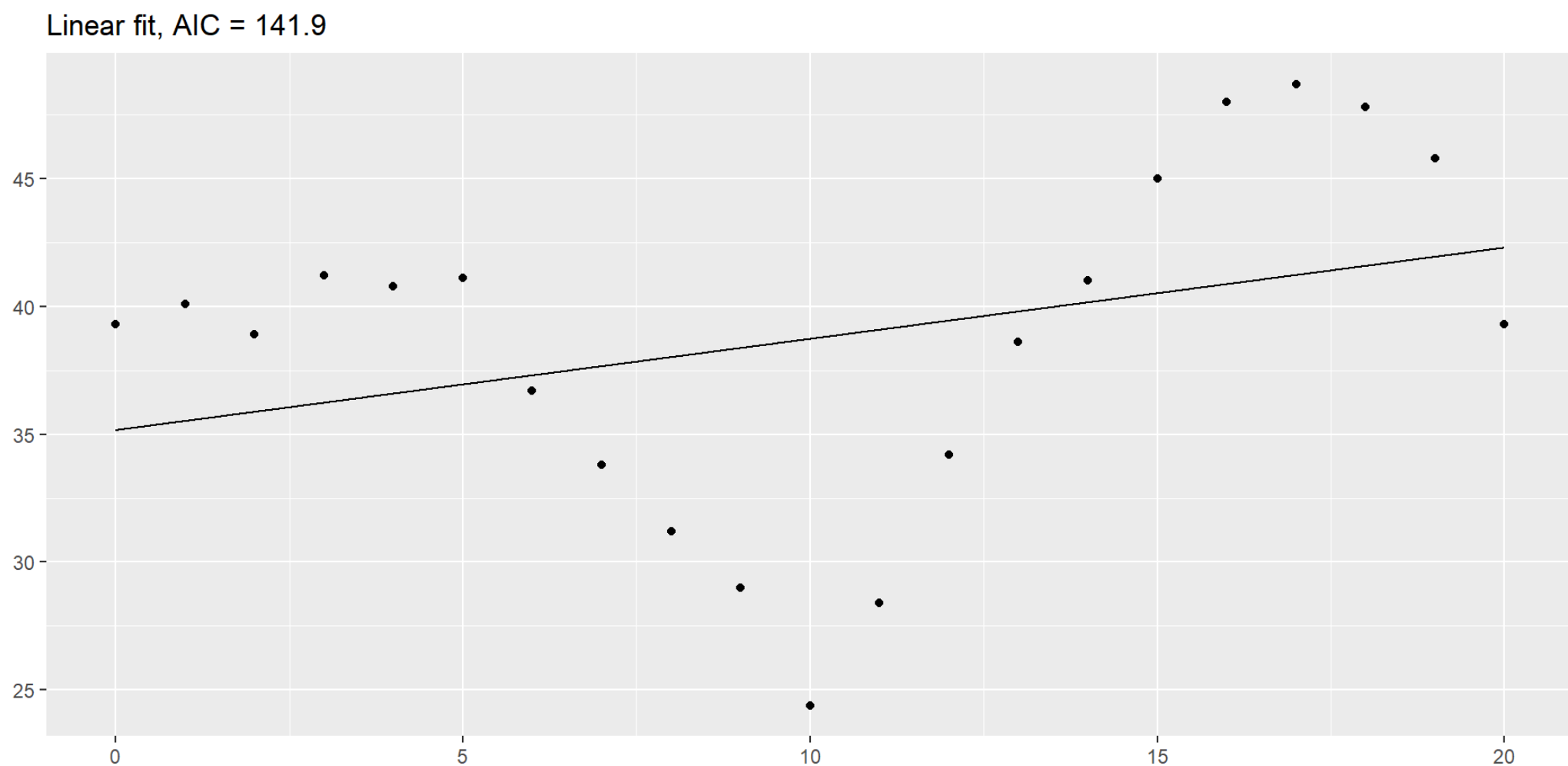
I am not an expert on cars, but I have been told that many newer cars with automatic transmission change how the transmission behaves around 40 miles per hour. This transition helps with highway mileage. So if you are fitting a spline curve to data where how the transmission behaves, make sure that you place one or two of your knots near 40 miles per hour.

I am also not an expert on kidneys, but I have been told that the Glomerural Filtration Rate is not too critical if the value is above 90, but becomes very serious when it is less than 30. So a model looking at health effects using GFR should probably have knots around 30 and 90.

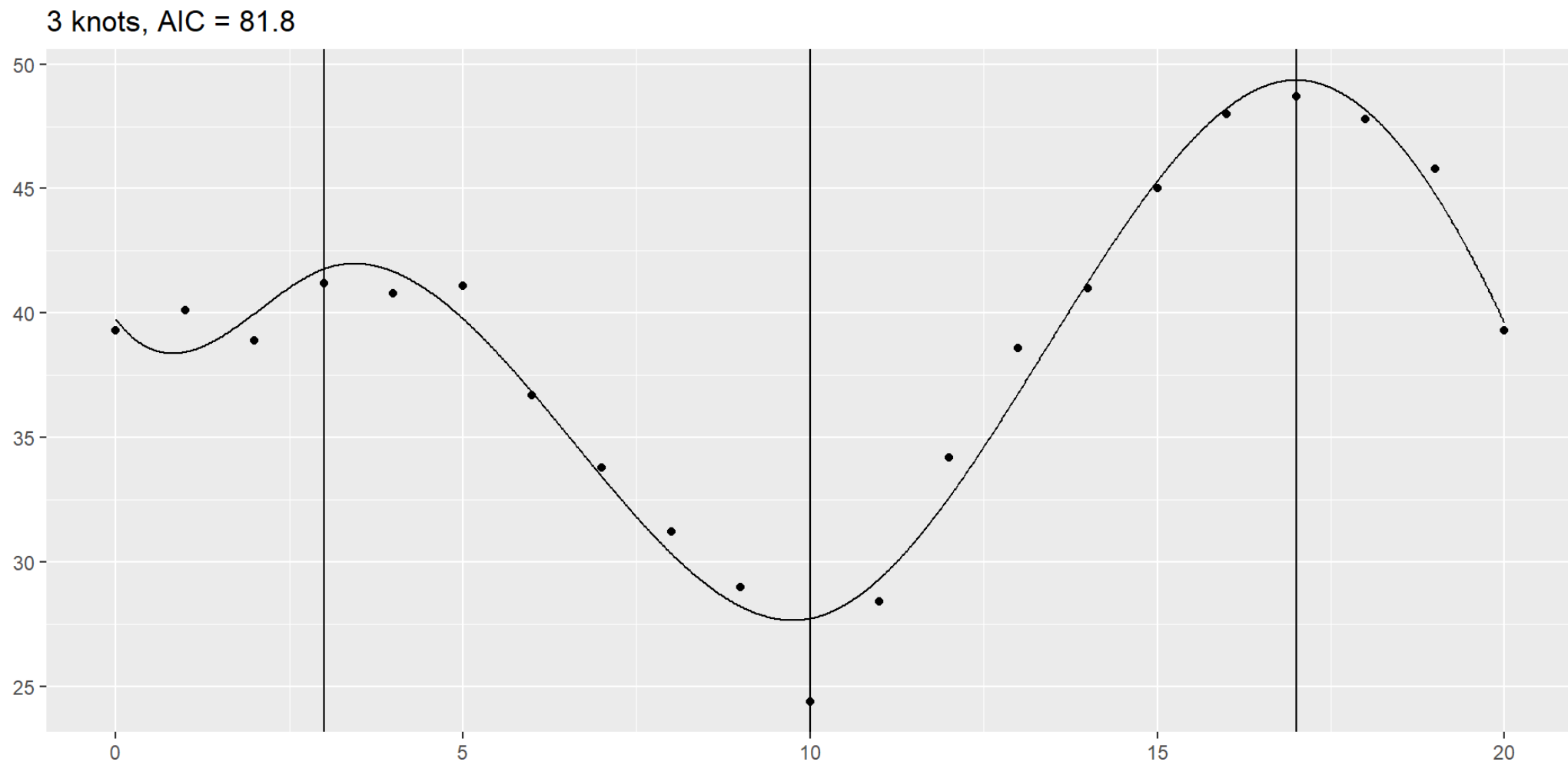
Similarly, CD4 cell counts above 500 are a good sign, but things turn rapidly worse if they dip below 200.

This is a bit controversial. Selecting a statistical model post hoc (after viewing the data) leaves you open to a charge of data dredging or going on a fishing expedition. To be fair, it is not as bad as some approaches (such as running ten tests and then choosing the one with the smallest p-value).

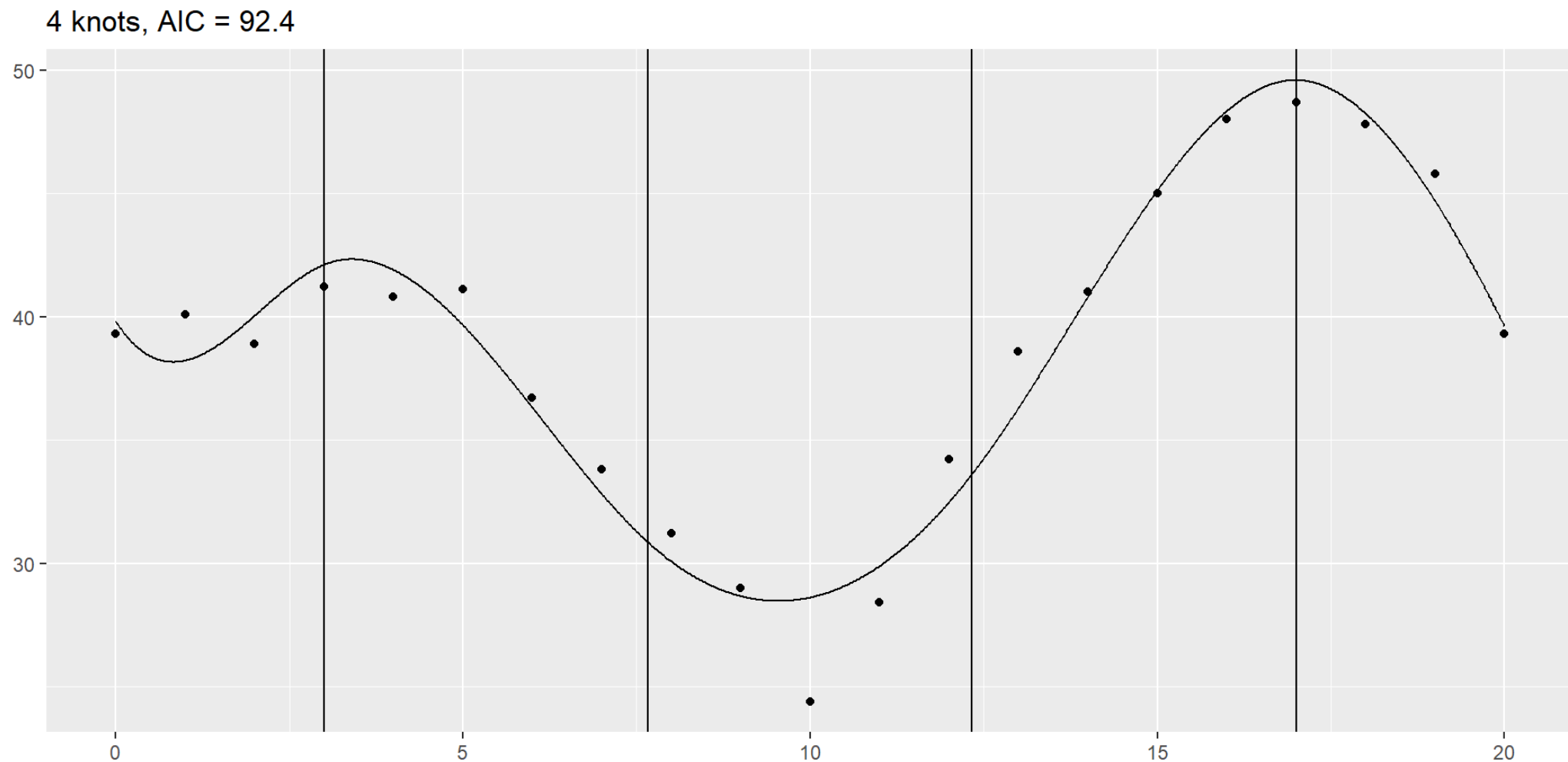
06-04 AIC for a linear fit



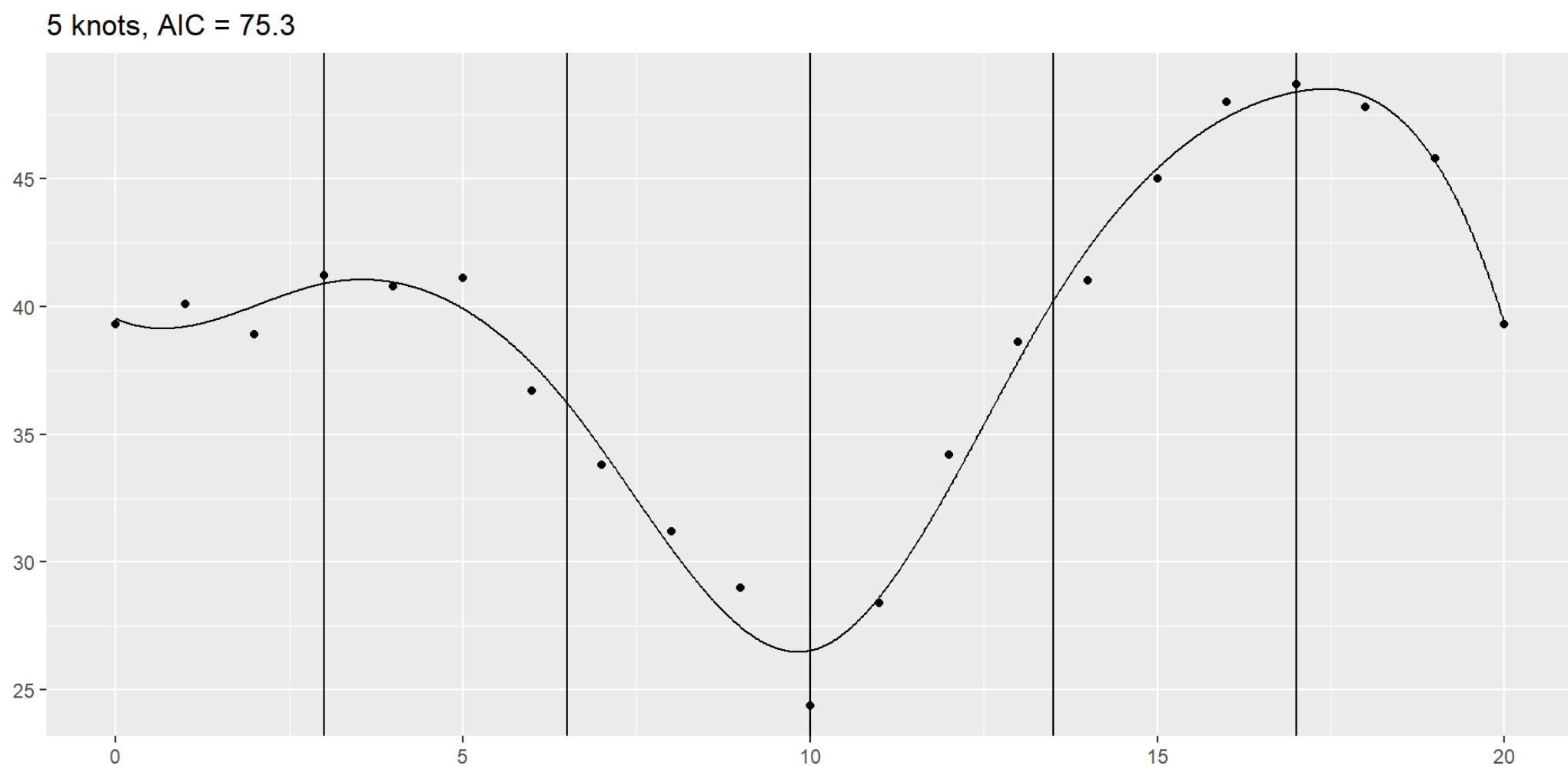
06-04 AIC for a three knot spline



06-04 AIC for a four knot spline



06-04 AIC for a five knot spline



06-04 Use inside knowlege to place the knots

- Automatic transmission locks in at 40 miles per hour
- Kidney function is fine above 90, problematic below 30
- CD4 counts fine above 500, trouble below 200

Speaker notes

I am not an expert on cars, but I have been told that many newer cars with automatic transmission change how the transmission behaves around 40 miles per hour. This transition helps with highway mileage. So if you are fitting a spline curve to data where how the transmission behaves, make sure that you place one or two of your knots near 40 miles per hour.

I am also not an expert on kidneys, but I have been told that the Glomerural Filtration Rate is not too critical if the value is above 90, but becomes very serious when it is less than 30. So a model looking at health effects using GFR should probably have knots around 30 and 90.

Similarly, CD4 cell counts above 500 are a good sign, but things turn rapidly worse if they dip below 200.

06-04 Frank Harrell’s suggestion for knot placement

- Evenly spaced percentiles
  - More knots where there is more data
- Set boundary knots close to edge
  - For 3 knots, use $q_{0.1}$ and $q_{0.9}$
  - For 4-6 knots, use $q_{0.05}$ and $q_{0.95}$
  - For 7 or more knots, use $q_{0.025}$ and $q_{0.975}$
- Example: 4 knots
  - Boundary knots at $q_{0.05}$ and $q_{0.95}$
  - Interior knots at $q_{0.35}$ and $q_{0.65}$

Speaker notes

Frank Harrell suggests that you place the knots not evenly across the range of X but at equally spaced quantiles of the X distribution. This makes sense when the distribution of the X values is not uniform. If, for example, X is skewed to the right (has a tendency to produce most of the data on the left with a few scattered outliers on the right), the knots will tend to favor the data-rich left side of the distribution. He also suggests placing the leftmost and rightmost knots near, but not at the extremes of the X values, such as at the 10th or 90th percentiles or at the fifth smallest and the fifth largest values in the data. The actual percentiles are a bit tricky to explain.

"For 3 knots, the outer quantiles used are 0.10 and 0.90. For 4-6 knots, the outer quantiles used are 0.05 and 0.95. For more than 6 knots, the outer quantiles are 0.025 and 0.975. The knots are equally spaced between these on the quantile scale. For fewer than 100 non-missing values of x, the outer knots are the 5th smallest and the 5th largest x." as quoted [here][har1],

If you pick this apart, you can deduce that 4 knots for a large dataset would be placed at the 5th, 35th, 65th and 95th percentiles.

06-04 Knot placement doesn’t matter

- Number of knots is important
- Placement of knots is not important

Speaker notes

Most references I have looked at state that it is the number of knots rather than the placement of the knots that is critical.

06-04 Extensions

- Natural splines
- Penalized splines
- More than one spline variable
  - Spline surfaces
  - Generalized additive models

Speaker notes

A variant on B splines are natural splines (also called restricted cubic splines). These splines place an additional restriction to the left of the first X value and to the right of the last X value. The spline is constrained to be linear at both extremes. This makes practical sense, as there is less data at the extremes, making estimation of a complex cubic function here worrisome. This also makes extrapolation outside of the range of data less problematic. Cubic polynomials have the potential of extreme shifts and if these occur outside the range of the data, they could lead to some awful extrapolations.

You should always be very careful, of course, as any effort to extrapolate beyond the range of data is [dangerous][sim3]. Nevertheless, restricting the extrapolation to a linear function is probably safer than letting the cubic polynomial wiggle around.

Penalized splines offer a way out if you are not sure how many knots to use. In a penalized spline, you put in 20 to 40 knots. This is way too many and could lead to overfitting. But then you compute a penalty term for excessive “wiggling”. It is effectively trying to minimize how much the spline coefficients change from one knot to the next.

How do you handle more than one spline variable? There are spline surfaces. The mathematics is tricky, but not terribly so. The problem is that surfaces can bend in many more ways than a line can. You can end up burning way too many degrees of freedom.

An alternative is to fit a spline for one variable and a spline for the other variables and just look at their additive effects. This is effectively the same as assuming that the splines do not interact with one another. This approach is called an additive spline model.

Break #4

- What you have learned
  - Placement and number of knots
- What's coming next
  - Splines for linear and logistic regression

06-05 About the Titanic

- Largest passenger ship at its time
- Maiden voyage in 1912
- Hit an iceberg and sank
- Detailed information on all passengers
  - Survived (no/yes)
  - Sex (female/male)
  - Passenger class (1st, 2nd, 3rd)
  - Age (2 months to 71 years)

Speaker notes

The Titanic was a large cruise ship, the biggest of its kind in 1912. It was thought to be unsinkable, but when it set sail from England to American in its maiden voyage, it struck an iceberg and sank, killing many of the passengers and crew. You can get fairly good data on the characteristics of passengers who died and compare them to those that survived. The data indicate a strong effect due to age and gender, representing a philosophy of “women and children first” that held during the boarding of life boats. Let’s look at the effect of age on survival using a logistic regression model.

06-05 First few rows of data

name	pclass	age	sex	survived
Allen, Miss Elisabeth Walton	1st	29	female	1
Allison, Miss Helen Loraine	1st	2	female	0
Allison, Mr Hudson Joshua Creighton	1st	30	male	0

06-05 A few descriptive statistics, 1

age_mean	age_sd	age_min	age_max	n
30.39799	14.25905	0.17	71	756

06-05 A few descriptive statistics, 2

pclass	n
1st	322
2nd	280
3rd	711

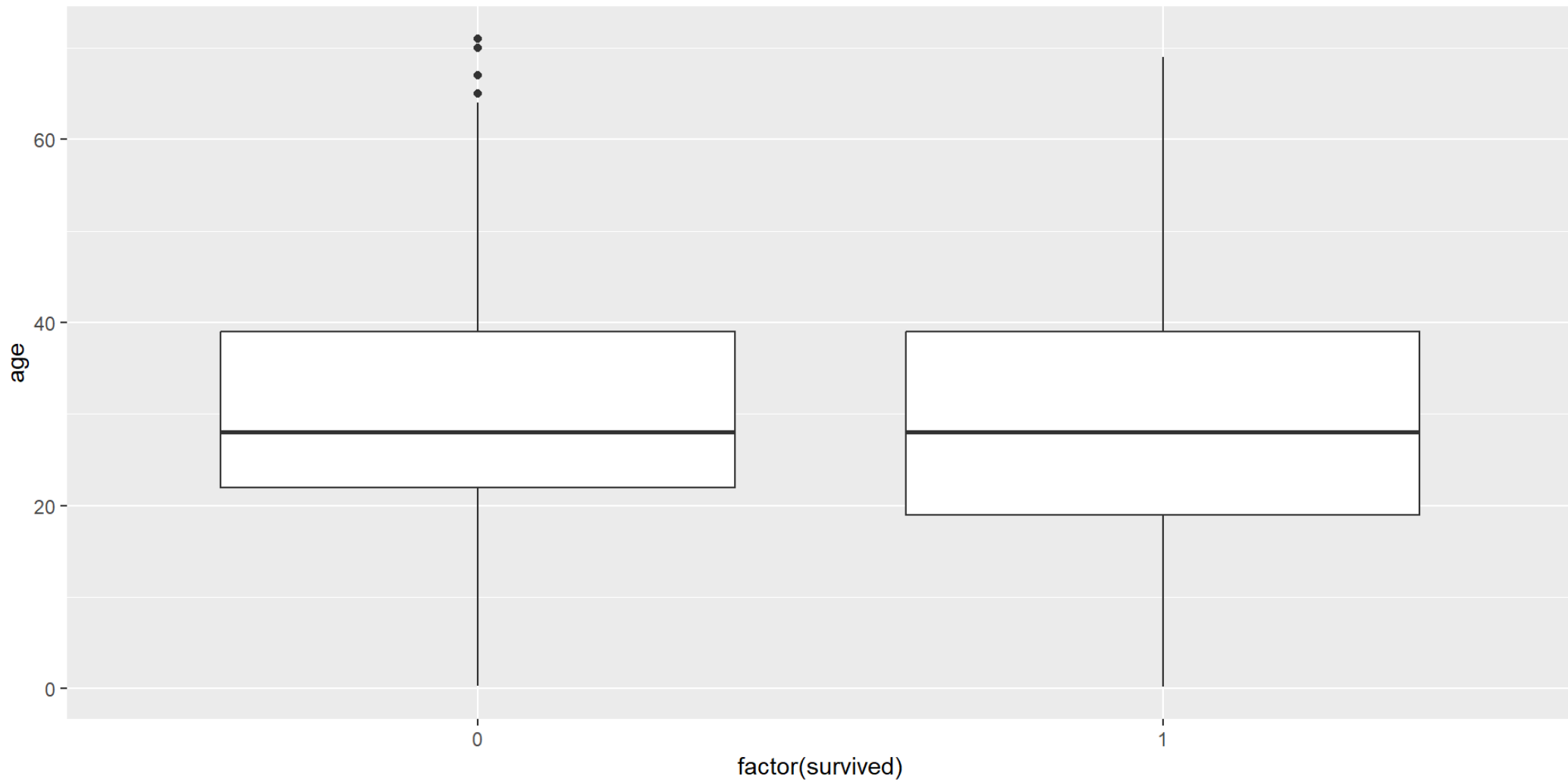
06-05 A few descriptive statistics, 3

sex	n
female	462
male	851

06-05 A few descriptive statistics, 4

survived	n
0	863
1	450

06-05 Boxplots



Speaker notes

The boxplots reveal little differences between the ages of survivors and deaths. If something is going on, it is subtle.

06-05 Fit a linear model first.

term	estimate	std.error	statistic	p.value
(Intercept)	-0.08142783	0.17386170	-0.4683483	0.63953556
age	-0.00879462	0.00523158	-1.6810637	0.09275054

Speaker notes

There may be a downward trend in the odds of survival over time, but it is not statistically significant.

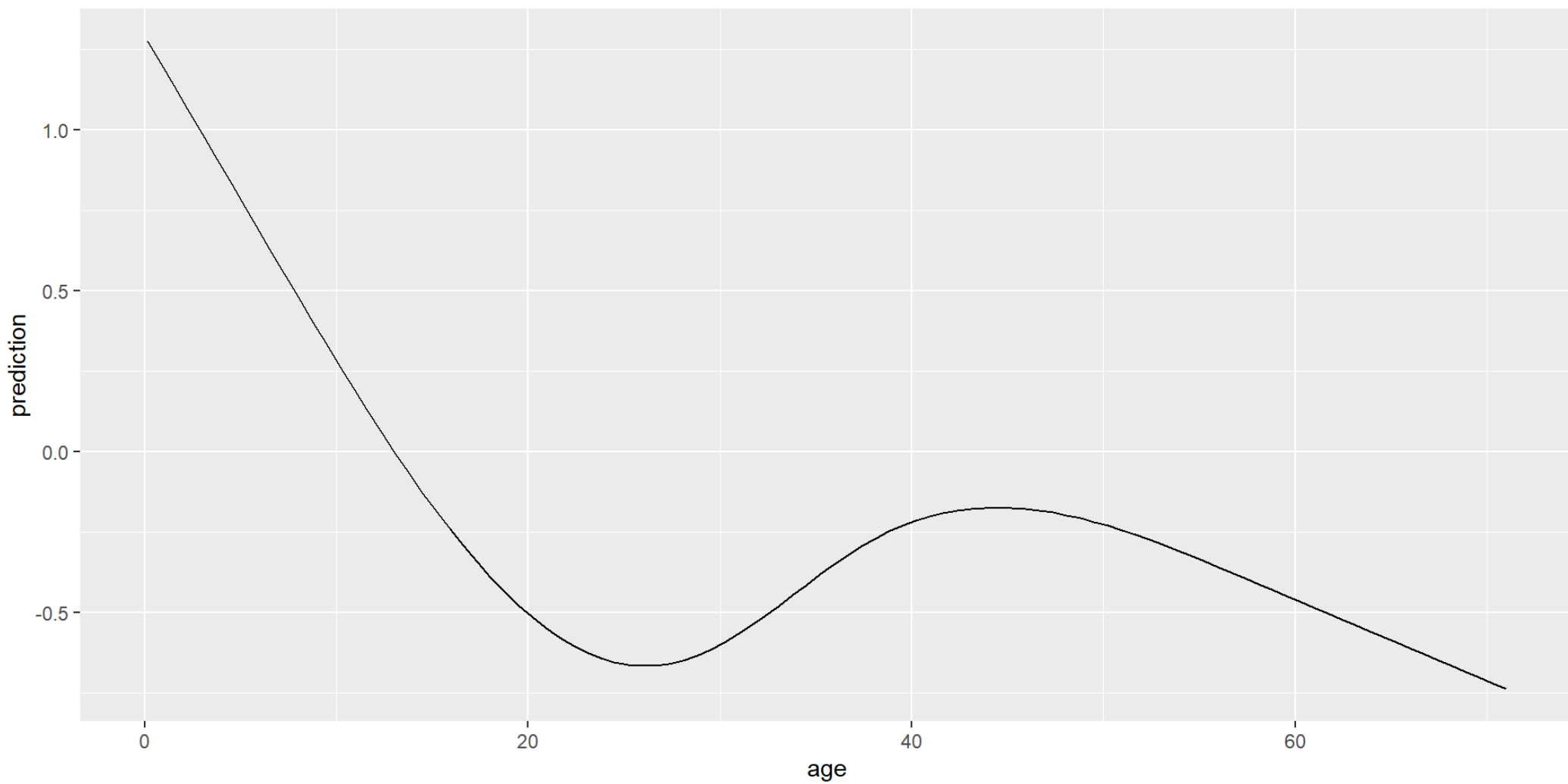
06-05 Now fit a spline function.

term	estimate	std.error	statistic	p.value
(Intercept)	1.2923278	0.39909550	3.2381418	0.0012031099
rcs(age)age	-0.1015127	0.02964437	-3.4243511	0.0006162695
rcs(age)age'	0.2225054	0.15885843	1.4006520	0.1613181579
rcs(age)age''	-0.1409403	1.19790026	-0.1176561	0.9063401205
rcs(age)age'''	-0.7520566	1.57492863	-0.4775179	0.6329933910

Speaker notes

The coefficients from the restricted cubic spline are pretty much uninterpretable. You have to visualize the spline graphically. First do this on the log odds scale to see how far from linear the spline fit is.

06-05 Predictions on a log odds scale

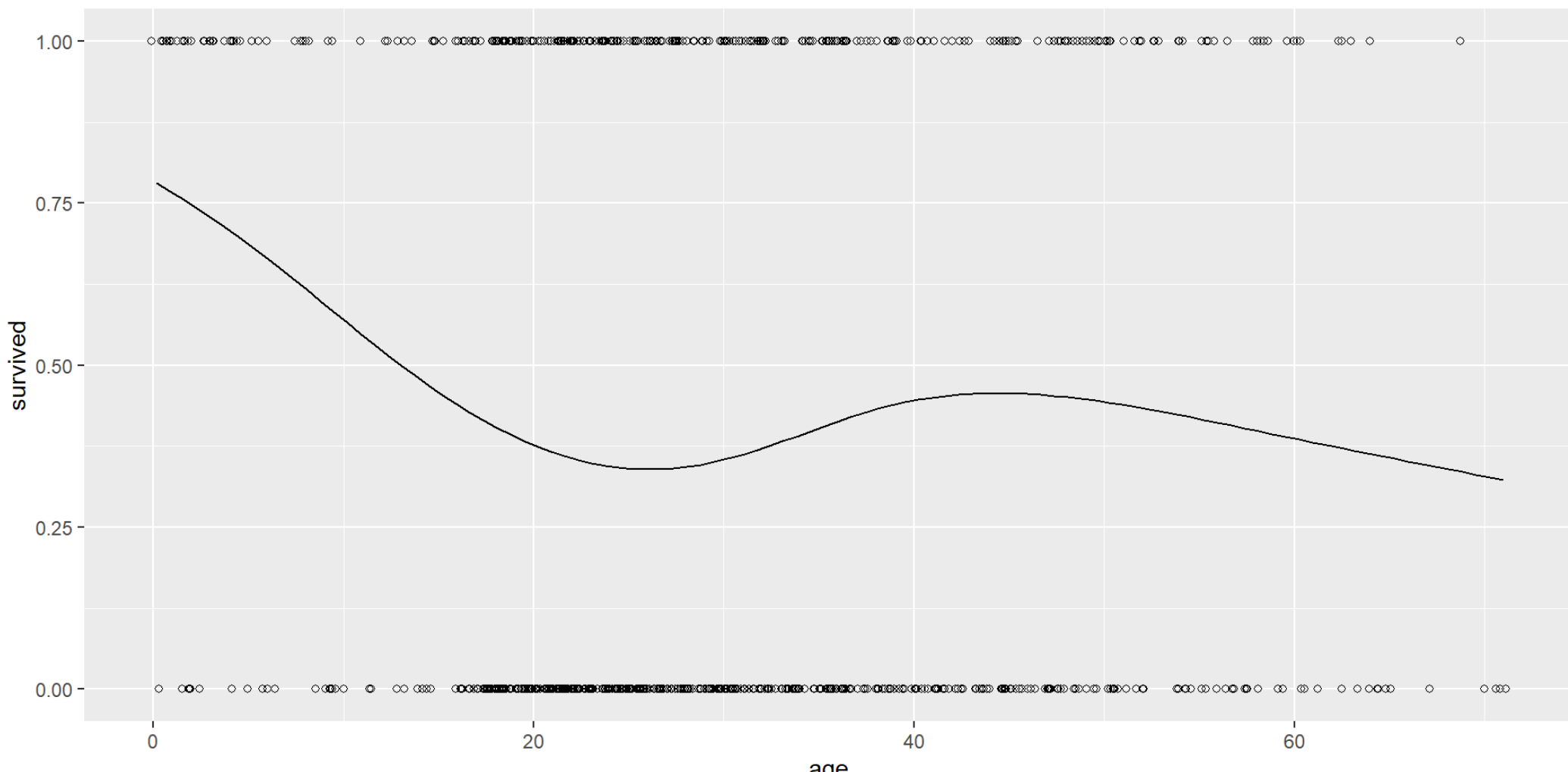


Speaker notes

It looks like a definite departure from linearity. The log odds for survival are best for very young children. They decline in older children and are at their worst for young adults. The log odds rise again starting at age 30 but this is a smaller and less dramatic change than you see in the children versus young adults.

The final dip for very old passengers, starting around 45, might be real and could reflect their relative frailty. But be careful about interpreting results in the tails.

06-05 Predictions on the probability scale



Speaker notes

The ages fall on whole number boundaries and in order to see things clearly, I jittered the data.

It looks like “women and children” first might actually be “women, children, and old people first”.

06-05 Most models will accept splines

- Easy
  - Linear regression
  - Random effects regression models
- Harder but not too difficult
  - Generalized linear model
  - Cox regression
- Not at all helpful for data science models
  - Random forests
  - Deep neural nets

Speaker notes

Splines work with a variety of statistical models. They work very nicely with linear regression and random effects regression models.

It takes a bit more work with the generalized linear model because of the nonlinear relationship with the outcome variable. With a bit of care, you will be fine.

Cox regression models also require a bit of care.

Many of the recent data science models do not work well with splines. The random forest model and deep neural nets, to name two, have their own way of modeling non-linearity and the spline functions would just get in the way.

Let me show how to use a spline in a logistic regression model with data from survival of passengers on the Titanic.

06-05 Spline coefficients are uninterpretable

term	estimate	std.error	statistic	p.value
(Intercept)	1.2923278	0.39909550	3.2381418	0.0012031099
rcs(age)age	-0.1015127	0.02964437	-3.4243511	0.0006162695
rcs(age)age'	0.2225054	0.15885843	1.4006520	0.1613181579
rcs(age)age''	-0.1409403	1.19790026	-0.1176561	0.9063401205
rcs(age)age'''	-0.7520566	1.57492863	-0.4775179	0.6329933910

Speaker notes

In the simple logistic regression model where age was treated linearly, you could exponentiate the coefficient to get an odds ratio. It turns out that the odds ratio for age is 0.99. Each additional year of age decreases the odds of survival by 1%. Not a lot, but it can add up across several decades of age.

You really can’t do anything with the spline coefficients. Go ahead and exponentiate them, but they won’t represent any meaningful odds ratio.

The ONLY way to interpret a spline model is through a graph.

06-05 Splines are a package deal

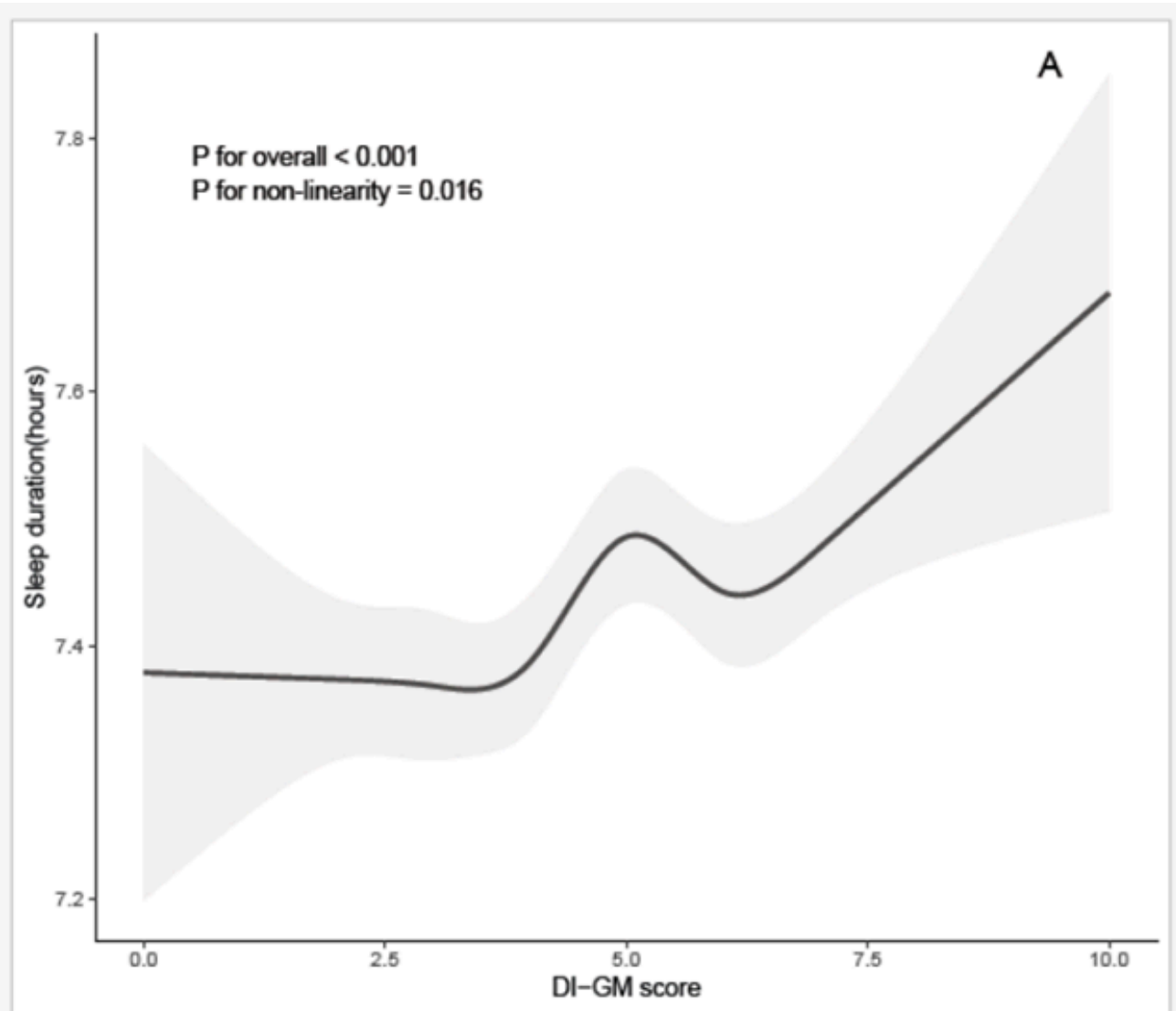
term	estimate	std.error	statistic	p.value
(Intercept)	1.2923278	0.39909550	3.2381418	0.0012031099
rcs(age)age	-0.1015127	0.02964437	-3.4243511	0.0006162695
rcs(age)age'	0.2225054	0.15885843	1.4006520	0.1613181579
rcs(age)age''	-0.1409403	1.19790026	-0.1176561	0.9063401205
rcs(age)age'''	-0.7520566	1.57492863	-0.4775179	0.6329933910

Speaker notes

The other thing to keep in mind is that the spline terms have to stick together. You can’t use stepwise regression to add or remove terms from the model and the individual p-values for the individual coefficients are misleading. You look at this output and you think to yourself that you can simplify the model by removing some of the spline terms with large p-values. You can’t do it. The spline coefficients are either all in or all out.

Now it is fairly easy to compare a spline model to a linear model. The linear model is implicitly nested within the spline model so you can just look at partial F tests, changes in R-squared, etc.

06-05 Don’t over-interpret the tails



Speaker notes

Standard errors are pretty easy to get with spline models. The mathematics behind this is tedious but not difficult. One thing that almost every spline exhibits is a fanning out of the standard errors at the extremes. This makes sense because there is less data to assess where the spline goes at the extremes.

So don’t get too worked up about unusual trends on the low end of the high end of the spline. It may be real or it may be sampling error.

Live demo, Splines for linear and logistic regression

Break #5

- What you have learned
  - Splines for linear and logistic regression
- What’s coming next
  - Your homework

Your homework

Summary

- What you have learned
  - A humorous look at nonlinear fits
  - Constructing a spline from small pieces
  - B-splines
  - Placement and number of knots
  - Splines for linear and logistic regression
  - Your homework